Knowledge-Aided Space-Time Adaptive Processing for Functional Ultrasound: Nuisance-Robust Microvascular Detection

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Abstract

Functional ultrasound (fUS) is increasingly used for vascular and functional mapping, but the interpretation of slow microvascular signal is often limited by nuisance rather than by average image quality: tissue motion, phase instability, and aliased high-velocity flow can generate rare false detections that disproportionately affect downstream maps. Low-rank clutter suppression methods such as motion-compensated singular value decomposition (MC–SVD) remain essential, but their residual outputs can still contain spatially varying interference in the low-false-positive-rate regime that matters most clinically.

We introduce a reduced-dimensional, post-beamformed space–time adaptive processing (STAP) detector for fUS slow-time data. The detector operates on localized tiles after a coarse clutter-suppression stage and produces a matched-subspace score for parenchymal Pf-band microvascular flow. An optional knowledge-aided (KA) spatial-spectral prior acts only as a conservative shrink-only penalty in high-risk regions; it never inflates scores and preserves a protected high-confidence set. We also study a conditional-execution variant explicitly as a changed decision rule, rather than a cost-free speedup.

We evaluate the method under a strict ultra-low-FPR operating discipline, reflecting the sensitivity of functional mapping to rare artifacts. Quantitative detection claims are restricted to regimes with independent structural labels: labeled k-Wave brain simulations, structurally labeled *in vitro* flow phantoms, and clinically anchored SIMUS/PyMUST moving-scatterer benchmarks. Baselines include MC–SVD, adaptive/global SVD, local and adaptive-local SVD, robust PCA, tensor HOSVD, and conventional short-ensemble Doppler heads. On the accepted clinically anchored structural benchmark, the best frozen deployed stack is RPCA→STAP, while a faster Adaptive Global SVD→STAP operating point preserves much of the nuisance-suppression benefit at substantially lower latency on an RTX 4080 SUPER. On the accepted functional benchmark, STAP is the stronger detector head in 11/12 family/profile comparisons under a common readout. Real-data retrospective audits show consistent nuisance-reduction gains on whole-brain fUS readouts and strongly positive same-residual detector-head results on ULM IQ bundles, whereas Shin real-IQ results remain mixed and are therefore treated as supportive rather than definitive. Overall, the results support STAP not as a replacement for upstream clutter suppression, but as a nuisance-focused detector head that can improve map trustworthiness when false positives matter clinically.

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1 Introduction

Functional ultrasound (fUS) is now used for vascular and functional mapping in settings ranging from preclinical neuroscience to awake surgery and cranial-window human imaging. In these settings, the practical problem is often not average sensitivity alone. A small number of nuisance-driven false detections can distort vascular readouts, contaminate functional maps, and undermine confidence in interpretation. Tissue motion, phase instability, and aliased high-velocity flow therefore matter most when they survive ordinary clutter suppression and populate the extreme right tail of the detector output.

We address that problem with a localized, STAP-inspired detector in which adaptation is temporal (slow-time), while spatial localization enters through tile-local covariance estimation. Instead of treating clutter suppression as the entire problem, we treat it as the first stage of a detection pipeline: a coarse residualizer removes obvious clutter, then STAP scores whether the residual is consistent with parenchymal microvascular flow. This targets the spatially varying interference patterns that dominate the low-FPR tail and often remain after otherwise reasonable low-rank filtering. Figure 1 provides a motivating qualitative example on a structurally labeled flow phantom: at a matched background-FPR operating point, STAP modifies detection decisions in a way that is consistent with the B-mode-derived vessel geometry.

Optionally, we add a conservative shrink-only regularization layer (knowledge aid) that compresses extreme-tail artifacts without score inflation: post-regularization scores are pointwise non-increasing, and an explicit do-not-penalize protected set is preserved (details in the Supplementary Material).

Contributions.

- A reduced-dimensional, post-beamformed STAP detector for fUS slow-time data: tile-local covariance adaptation in slow time with a matched-subspace Pf-band score, evaluated where low false positives matter clinically.
- A benchmark discipline that separates upstream residualizers from detector heads, includes strong low-rank and tensor baselines, and evaluates performance under strict low-FPR operating points rather than image-quality heuristics alone.
- Reproducible latency and deployment analyses that treat conditional execution explicitly as a changed decision rule rather than an unconditional speedup.
- A clinically anchored SIMUS/PyMUST benchmark track and a compact real-data translation audit that together support a nuisance-focused interpretation of STAP rather than a claim of universal superiority in every regime.

Scope and claims. Claims are stratified by data regime and summarized in Table 2. Quantitative detection efficacy is limited to regimes with independent H_1/H_0 labels; unlabeled in-vivo IQ and PD-only datasets are used for instrumentation and robustness analyses.

Organization. Section 2 defines the detection problem and score taxonomy. Section 3 presents the localized detector and summarizes the optional shrink-only KA regularization (full KA machinery in the Supplementary Material). Section 4 specifies fixed-profile experimental design and integrity controls. Section 5 reports results, followed by discussion and conclusion.

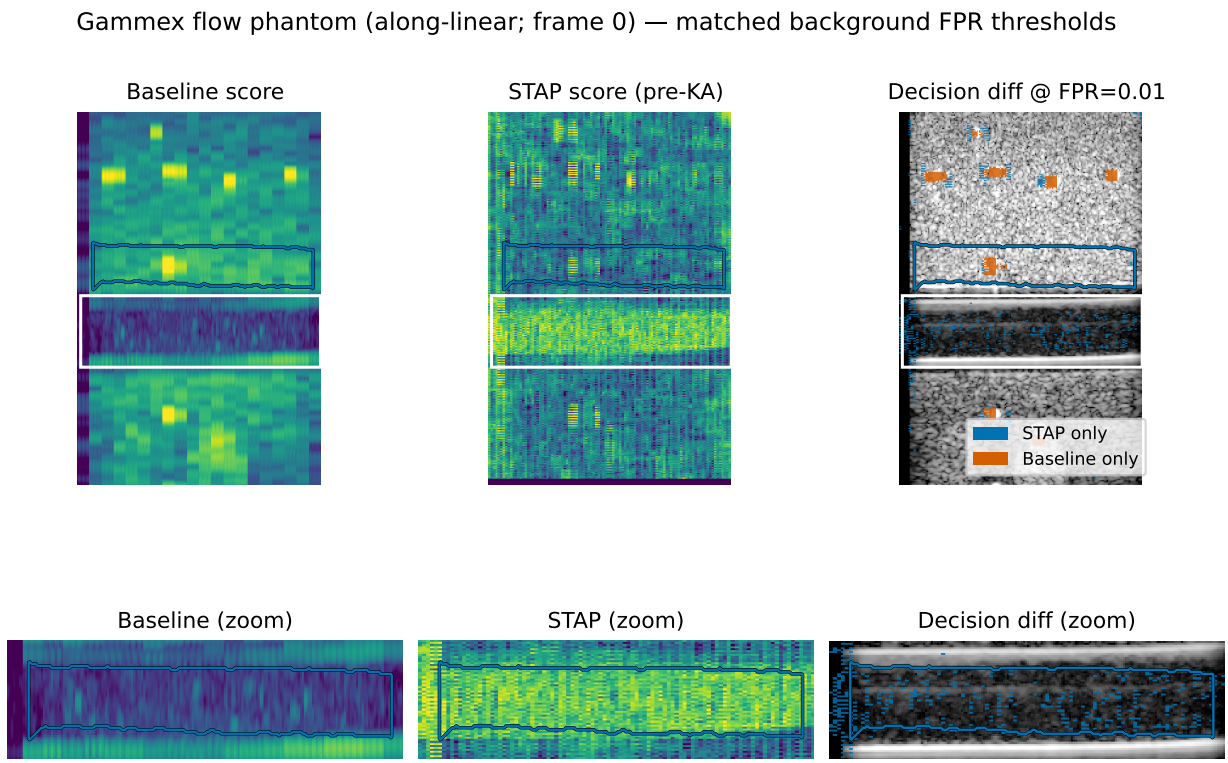

Figure 1: Leading structural-fidelity visual on a structurally labeled Gammex flow phantom acquisition (labels derived from B-mode-only tube segmentation). Top row: baseline score map S_{base} (SVD band-pass clutter suppression followed by power Doppler), STAP pre-KA matched-subspace score $S_{\text{stap,pre}}$, and a matched-FPR decision-difference panel at a fixed background false-positive rate (10^{-2}; chosen here for visualization), showing pixels detected by STAP only versus baseline only. Bottom row: zoomed region of interest. The tube contour is overlaid for reference.

2 Background and Problem Formulation

Problem setup. For each spatial location (or tile-aggregated snapshot), let $x \in \mathbb{C}^{L_t}$ denote a slow-time vector. We consider

$$H_0 : \; x = c + n, \qquad H_1 : \; x = s + c + n,$$

where s is flow-consistent signal, c is structured clutter/motion, and n is residual noise. In practice, H_0 is spatially nonstationary and heavy-tailed, motivating local covariance adaptation and tail-focused evaluation.

Core notation and bands. We use band operators (P_o, P_f, P_g, P_a) for clutter/DC, flow, guard, and alias regions, respectively. For short ensembles, these are exact orthogonal DFT-bin projectors; for longer apertures they are leakage-controlled analysis operators and are treated as band-energy statistics with explicit sanity sentinels (Supplementary Material).

2.1 Related context

STAP and knowledge-aided STAP. Classical MVDR/LCMV/STAP and knowledge-aided variants shape covariances using external structure [50–53, 58, 59, 62]. In radar, KA is often expressed as covariance shaping/blending or knowledge-aided covariance estimation; here we adopt the KA spirit but implement it as a shrink-only, fail-safe score regularizer on top of a fixed detector statistic, with explicit conditions for inert behavior out of regime.

Ultrasound clutter suppression. Spatiotemporal SVD clutter filters are a standard tool in ultrafast Doppler and fUS [6, 7, 11, 12, 18]. Real-time feasibility has been studied via randomized/approximate decompositions and downsampling strategies [17], as well as GPU-parallel eigen-based clutter filtering frameworks [16]. Robust PCA, unfolded/learned RPCA, and tensor/HOSVD variants have also been studied for clutter suppression and motion robustness [24,25,42]. More generally, when raw channel/aperture data are available, aperture-domain multidimensional clutter filtering can provide additional degrees of freedom beyond beamformed slow-time processing [41]. We benchmark STAP against MC–SVD and established SVD-family clutter filters spanning adaptive-global SVD (spatial singular-vector similarity cutoffs), block-wise local SVD with frozen tilewise energy thresholds, and an adaptive-local hybrid that applies the same similarity-drop rule tilewise [12, 14], as well as RPCA and HOSVD baselines; conventional short-ensemble Doppler comparators are additionally reported on phantom data. These baselines are included to represent state-of-practice eigen/SVD clutter filtering beyond fixed-cutoff global Casorati SVD and to preempt the common critique that strict-tail gains are attributable only to localization or cutoff tuning rather than to the detector formulation and tail operating discipline.

Relation to eigen-based and local adaptive clutter filtering. Eigen-based clutter filter design for ultrasound color flow imaging and block-wise/local adaptive clutter filters are closely related ultrasound-native approaches to covariance/eigenspace adaptation [13, 14, 16]. Song *et al.* demonstrate local clutter filtering in *in vivo* settings (e.g., kidney imaging), while Baranger *et al.* provide an adaptive singular-vector-similarity rank rule for ultrafast Doppler [12]. Our benchmark set includes both literature families directly and a tile-local hybrid that combines Song-style overlap-add localization with a Baranger-style adaptive rank rule. Many open fUS releases do not preserve raw high-PRF complex IQ suitable for labeled Doppler-side detector evaluation, so we use labeled k-Wave pilots and structurally labeled phantoms for quantitative ROC claims and use *in vivo* data for robustness and instrumentation audits. Our contribution differs in the *detector* formulation and evaluation discipline: in contrast to approaches framed primarily as clutter suppression followed by power Doppler scoring, we compute a GLRT-motivated matched-subspace detector score on tile-local slow-time snapshots with explicit Doppler band geometry (flow/guard/alias) and optional tile-local whitening, evaluate at empirically supported ultra-low FPR operating points where background-tail behavior dominates, and treat compute gating (conditional execution) as a distinct decision rule with leakage controls.

ULM pipelines and motion correction. Ultrafast ultrasound localization microscopy (ULM) achieves deep super-resolution vascular imaging by localizing sparse microbubble echoes over high-PRF ensembles [38]. Robust clutter suppression and motion correction are central to ULM pipelines, including subwavelength motion correction tailored to ultrafast acquisitions [39]. We do not claim improved microbubble localization performance here; rather, we use a public kHz IQ ULM benchmark dataset to report label-free stability and motion robustness of Doppler-side score maps under controlled motion corruption, illustrating how a tail-stable detector can complement upstream motion correction and downstream localization stages.

Twinkling artifact and Doppler phantoms. The Doppler "twinkling artifact" has been documented as a persistent color-flow phenomenon on calcifications and other strong reflectors, and is widely used as a diagnostic benchmark for Doppler processing chains. Leonov *et al.* provide an open-access database of raw beamformer signals with phantoms including a flow-through Gammex phantom and calculi/metal/plastic objects [40]. We use these data to (i) obtain a structurally labeled phantom ROC via B-mode-only tube segmentation and (ii) construct an artifact-heavy KA hygiene regime where the prior penalty is expected to be conservative and auditable.

fUS detection and hemodynamic analysis. GLM-based detection and atlas-driven ROI analysis are common in hemodynamic fUS [45,46]; our hemodynamic STAP variant reuses the same banded temporal machinery on PD time series and adds an alias-aware shrink-only prior penalty as a safety layer.

2.2 Baseline clutter suppression and score conventions

Beamformed slow-time data. Let $X[t, y, x] \in \mathbb{C}$ denote a beamformed slow-time IQ movie over $t = 0, \ldots, T-1$ and spatial indices (y, x). We first apply a baseline clutter-suppression operator (e.g. motion-compensated SVD), producing a residual cube $X_{\text{base}}[t, y, x]$ used for both baseline scoring and as the input to STAP.

Power Doppler proxy and baseline scores. From the baseline-filtered cube we define a conventional power Doppler (PD) brightness map used only as a display/proxy map for deterministic preprocessing (e.g. proxy masks) and visualization,

$$\mathrm{PD}^{\uparrow}(y, x) = \frac{1}{T} \sum_{t=0}^{T-1} |X_{\text{base}}(t, y, x)|^2 .$$

For quantitative evaluation and thresholding we use *flow-evidence* score maps where higher values indicate stronger evidence of flow, and we calibrate thresholds on the right tail of these score distributions. We denote the baseline ROC score map as $S_{\text{base}}(y, x)$ (power Doppler on X_{base}). For short-ensemble regimes we additionally report auxiliary, conventional baseline comparators (e.g. log-power Doppler and Kasai lag-1 magnitude [15]) to avoid baseline-score idiosyncrasies dominating conclusions (Notation). Note: $\mathrm{PD}^{\uparrow}$ and S_{base} use the same underlying power statistic (up to fixed scaling); we keep separate names to make it explicit when PD is used only as a display/proxy map (masks/visuals) versus as a ROC score map.

2.3 Score taxonomy

Unless explicitly stated, "STAP" refers to the pre-KA matched-subspace score $S_{\text{stap,pre}}$; KA is analyzed as an optional, shrink-only post-processing regularizer.

3 Method

Algorithm 1 (STAP pipeline with optional KA regularization, fixed profile).

1. Baseline clutter suppression (e.g. MC–SVD) produces a residual cube X_{base} and baseline score $S_{\text{base}}(y, x)$.

Symbol / name	What it measures	Used for	Data requirement
S_{base}	baseline ROC score map on X_{base} (PD power; higher = more flow-like)	baseline ROC / low-FPR tables	IQ
$S_{\text{stap,pre}}$	pre-KA matched-subspace ratio STAP score map	STAP ROC / low-FPR tables	IQ + local covariance
$S_{\text{stap,post}}$	post-KA shrink-only score map $wS_{\text{stap,pre}}$	KA behavior audits	IQ + feature map
$\text{PD}^{\uparrow}$ (display PD), masks	PD display/proxy map and derived masks	visualization and deterministic preprocessing (gating/protected sets)	IQ (or PD-only proxy mode)

Table 1: Score/map taxonomy used throughout. ROC score maps are kept separate from display/proxy maps used only for visualization and deterministic preprocessing. Display/proxy maps are never used to compute ROC/TPR/FPR (only to define gating/protected-set masks, which are audited separately).

2. For each tile Ω_i: form slow-time snapshots $\{x_{i,k}\}_{k=1}^{K}$ via spatial sampling and, optionally, a sliding-window temporal embedding (Hankel matrix construction), estimate conditioned covariance R_i, and whiten $y_{i,k} = R_i^{-1/2} x_{i,k}$.
3. Compute a matched-subspace ratio score (ACE/MSD-style) on whitened snapshots, e.g. $S_{\text{stap,pre}}(i) = \frac{1}{K}\sum_k \frac{y_{i,k}^{\text{H}} P_f\, y_{i,k}}{y_{i,k}^{\text{H}}(I-P_f)\, y_{i,k} + \rho\, y_{i,k}^{\text{H}} y_{i,k} + \epsilon}$, and aggregate over overlapping tiles to obtain $S_{\text{stap,pre}}(y,x)$.
4. If KA enabled: compute a deterministic, shrink-only weight $w(i) \in (0,1]$ from auxiliary band-energy statistics and proxy-support features (details in the Supplementary Material), preserve a protected do-not-penalize set (scores unchanged), and output $S_{\text{stap,post}}(y,x) = w\, S_{\text{stap,pre}}(y,x)$. Else, $S_{\text{stap,post}}(y,x) = S_{\text{stap,pre}}(y,x)$.

In practice STAP is intended as a second-stage residual covariance whitener: a coarse baseline clutter filter removes gross stationary tissue clutter, and STAP targets the remaining structured, spatially varying interference that dominates extreme-tail false alarms. A STAP-only ablation on phantom data is provided in Supplementary Material.

3.1 Temporal STAP core and Doppler band geometry

Tile-based STAP. To mitigate the degraded tail sensitivity of global low-rank filters in high-clutter regimes, we apply a localized temporal STAP core on overlapping spatial tiles. We use "STAP" in a STAP-inspired sense: the adaptive filter operates in slow time, while spatial localization enters through tile-local covariance adaptation. Let Ω_i denote tile i and L_t the slow-time aperture. From X_{base} we construct a set of slow-time training snapshots $\{x_{i,k} \in \mathbb{C}^{L_t}\}_{k=1}^{K}$ for each tile via spatial sampling and, when applicable, a sliding-window temporal embedding (Hankel construction) in slow time. Here K denotes the number of training snapshots used to estimate the tile covariance; in the fully Hankelized case, each pixel contributes $N = T - L_t + 1$ snapshots and pooling across Ω_i yields $K = N|\Omega_i|$ (or a subsampled/capped variant). We estimate a local

covariance

$$\widehat{R}_i := \frac{1}{K}\sum_{k=1}^{K} x_{i,k}x_{i,k}^{\mathrm{H}},$$

apply conditioning/diagonal loading to obtain R_i, and whiten snapshots as $y_{i,k} := R_i^{-1/2}x_{i,k}$.

Training support and Hankel dependence. In the fully Hankelized case, each pixel in Ω_i contributes $N = T - L_t + 1$ overlapping slow-time snapshots; pooling across the $|\Omega_i|$ pixels yields $K = N|\Omega_i|$ tile-local training vectors. These Hankelized snapshots are temporally correlated due to overlap, so K should not be interpreted as an independent-sample count. Our robust covariance estimators do not require independence, but the effective support can be significantly smaller than K; we therefore optionally subsample Hankel columns (stride) and cap the number of columns used for training to bound computation and avoid overweighting highly correlated samples (fixed per regime; Supplementary Material).

Robust covariance estimation and conditioning. We whiten with a regularized covariance R_i rather than the raw SCM $\widehat{R}_i$. In the primary profile we use a Tyler-type M-estimator on the pooled tile snapshots, iterating

$$R_i^{(n+1)} \leftarrow \frac{L_t}{K}\sum_{k=1}^{K}\frac{x_{i,k}x_{i,k}^{\mathrm{H}}}{x_{i,k}^{\mathrm{H}}(R_i^{(n)})^{-1}x_{i,k}}, \qquad R_i^{(n+1)} \leftarrow \frac{L_t}{\mathrm{tr}(R_i^{(n+1)})}\,R_i^{(n+1)},$$

with at most 25 iterations and early stopping when $\|R^{(n+1)} - R^{(n)}\|_F/\|R^{(n)}\|_F < 10^{-4}$ (latency-only experiments may cap iterations; Supplementary Material). We then apply deterministic conditioning (shrinkage toward μI and diagonal loading $\lambda\mu I$, where $\mu = \mathrm{tr}(R)/L_t$ and λ is fixed per regime) to enforce target condition numbers. In very short-ensemble regimes we may fall back to a Huber IRLS covariance estimate for numerical stability (Supplementary Material).

Terminology: "space–time" in fUS STAP. In classical radar STAP, a snapshot stacks spatial array channels and temporal pulse taps, and adaptation uses training snapshots from nearby range bins [50, 62]. In beamformed fUS IQ we operate on a single complex slow-time sample per pixel, so we do not exploit array-manifold degrees of freedom; our space–time vector is the slow-time stack $x \in \mathbb{C}^{L_t}$, while "space" enters through *localized adaptation*: we estimate R_i from snapshots drawn from a small spatial neighborhood (tile) assumed approximately stationary over that region. This yields a reduced-dimensional, post-Doppler STAP-like detector: we whiten by $R_i^{-1/2}$ and score energy in an explicitly defined Doppler flow subspace using a matched-subspace statistic (Section 3), rather than framing the method solely as clutter suppression followed by power Doppler. We use the term "STAP" to emphasize these defining elements (local covariance adaptation, whitening, and subspace detection) while being explicit that channel-domain STAP/beamforming is out of scope for the present beamformed pipeline.

Figure 2 illustrates the local tile extraction and optional sliding-window temporal embedding (Hankel construction) used to form $\widehat{R}_i$ and contrasts it with the global Casorati stacking used by MC–SVD.

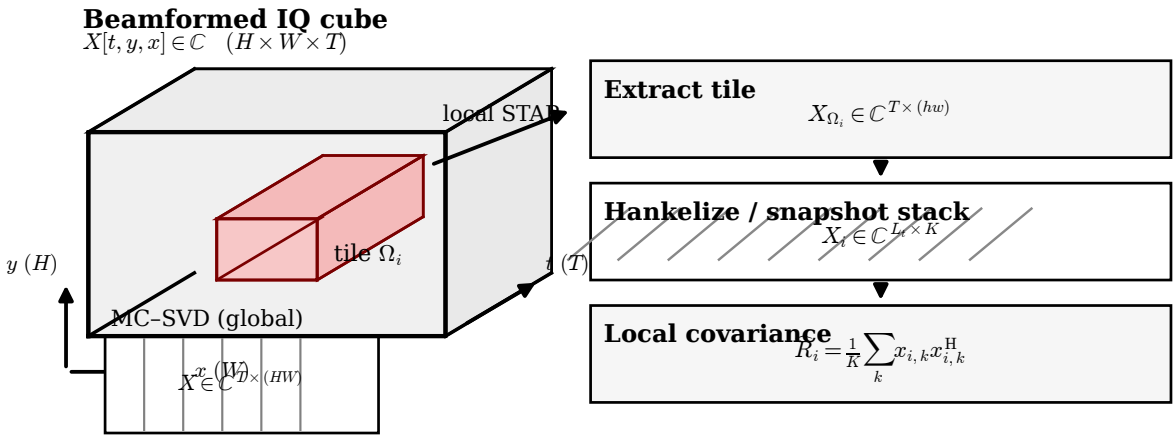

Figure 2: Schematic of sliding-window temporal embedding (Hankel matrix construction) and local covariance estimation. A localized spatial tile Ω_i is extracted from the beamformed slow-time data cube $X[t, y, x] \in \mathbb{C}$ $(H \times W \times T)$. Overlapping slow-time snapshots of length L_t are stacked (optionally Hankelized) to estimate the localized covariance matrix $\widehat{R}_i$. Unlike global low-rank filters (e.g. MC–SVD) that operate on a global Casorati matrix, this localized formulation preserves spatial-spectral heterogeneity of microvascular flow and highly structured tissue clutter.

Band operators and matched-subspace STAP score. We partition the Doppler spectrum using band projectors (P_o, P_f, P_g, P_a) (Notation). For short ensembles, these are exact, orthogonal DFT-bin projectors with $P_o + P_f + P_g + P_a = I$; for longer apertures they are leakage-controlled operators that are approximately orthogonal and are treated as band-energy statistics. We define a matched-subspace STAP test statistic by flow-subspace energy on whitened snapshots, e.g.,

$$S_{\mathrm{stap,pre}}(i) := \frac{1}{K} \sum_{k=1}^{K} \frac{y_{i,k}^{\mathrm{H}} P_f \, y_{i,k}}{y_{i,k}^{\mathrm{H}} (I - P_f) \, y_{i,k} + \rho \, y_{i,k}^{\mathrm{H}} y_{i,k} + \epsilon},$$

and form a pixel-level score map $S_{\mathrm{stap,pre}}(y, x)$ by overlap-add aggregation of the contributing tile statistics. This pre-KA score is the primary STAP detector output used for quantitative reporting unless explicitly stated otherwise.

Regularization and self-normalization avoidance. If we whitened using the exact sample covariance formed from the same snapshots being scored, the statistic collapses: the whitened sample covariance becomes (approximately) identity and averaged subspace energies concentrate near constants (e.g. $\mathrm{tr}(P_f)$), destroying discrimination. We avoid this by whitening with a *regularized* covariance R_i (robust estimation plus shrinkage/diagonal loading), so that typically $R_i \neq \widehat{R}_i$ and the matched-subspace energy remains informative. We do not rely on an analytic null distribution for CFAR operation; operating points are empirically calibrated on labeled H_0 sets under the fixed-profile protocol of Section 4.1.

Matched-subspace statistic as a whitened subspace detector. Fix tile i and consider a single slow-time snapshot $x \in \mathbb{C}^{L_t}$. Under a Gaussian clutter-plus-noise model,

$$H_0: \; x \sim \mathcal{CN}(0, R_i), \qquad H_1: \; x = A_f s + n, \quad s \sim \mathcal{CN}(0, \sigma_s^2 I), \quad n \sim \mathcal{CN}(0, R_i),$$

where the columns of A_f span the flow subspace (equivalently the range of P_f). After whitening $y = R_i^{-1/2} x$, the GLRT for $\sigma_s^2 > 0$ versus $\sigma_s^2 = 0$ is monotone with the whitened subspace energy

$$T(x) = x^{\mathrm{H}} R_i^{-1/2} P_f R_i^{-1/2} x = y^{\mathrm{H}} P_f \, y,$$

which is a standard matched-subspace / adaptive energy detector. [60, 61] In implementation we report a self-normalized ACE/MSD-style ratio score that normalizes flow-subspace energy by the orthogonal complement (and a small ridge), which is monotone in $T(x)$ and serves as our pre-KA detector score $S_{\text{stap,pre}}(i)$.

3.2 Optional shrink-only tail regularization (KA safety layer)

While the matched-subspace statistic $S_{\text{stap,pre}}$ effectively isolates flow under standard colored-interference models, structured artifacts can still dominate extreme-tail operating points. We therefore study an *optional* knowledge-aided (KA) layer that applies a conservative, shrink-only score penalty in high-risk regions. The KA mapping is deterministic and uses auxiliary band-energy and proxy-support statistics only as *side information*; proxies are never used to compute ROC/TPR/FPR (only to define protected/candidate sets and compute masks, which are audited separately).

KA in one picture (at-a-glance summary).

- **When it activates:** Only when deterministic sanity/support checks pass (valid band discretization; non-flat risk features; sufficient proxy support). Otherwise it is inert ($w \equiv 1$).
- **What it changes:** Applies a bounded, shrink-only weight $w(i) \in (0, 1]$ on an explicitly defined candidate set while preserving a protected do-not-penalize set (scores unchanged); the post-KA score is $S_{\text{stap,post}} = w \, S_{\text{stap,pre}}$.
- **What it cannot do:** It cannot increase any pixel score (no amplification) and proxies are not used for ROC/TPR/FPR reporting; any apparent operating-point uplift reflects background-tail compression under shrink-only mapping.
- **How it fails safely:** If out of regime, it defaults to "KA off" ($w \equiv 1$) and logs an explicit reason label, reducing to the pre-KA STAP detector.

Shrink-only mapping (guarantee). We implement knowledge aid via a bounded weight $w(i) \in (0, 1]$ applied in score space:

$$S_{\text{stap,post}}(i) = w(i) \, S_{\text{stap,pre}}(i), \qquad w(i) \in (0, 1].$$

Equivalently, define the per-tile scale $\gamma(i) := 1/w(i) \geq 1$ and write $S_{\text{stap,post}}(i) = S_{\text{stap,pre}}(i)/\gamma(i)$. The construction of $w(i)$ (feature definitions, proxy sets, regimes, and deterministic sentinels) is provided in Supplementary Material and Supplementary Material.

Lemma 1 (Shrink-only monotonicity)**.** *Let* $w(i) \in (0, 1]$ *and define* $S_{\text{stap,post}}(i) = w(i)\, S_{\text{stap,pre}}(i)$*. For any threshold* τ*,*

$$\{S_{\text{stap,post}} \geq \tau\} \subseteq \{S_{\text{stap,pre}} \geq \tau\}.$$

Consequently, on any subset A *(e.g.* H_0*), the tail probability cannot increase at fixed* τ*:*

$$\mathbb{P}\{S_{\text{stap,post}} \geq \tau \mid A\} \leq \mathbb{P}\{S_{\text{stap,pre}} \geq \tau \mid A\}.$$

Proof. If $S_{\text{stap,post}}(i) \geq \tau$, then $S_{\text{stap,pre}}(i) = S_{\text{stap,post}}(i)/w(i) \geq \tau$ since $w(i) \leq 1$. □

When reporting strict low-FPR operating points where τ is reselected post-KA to satisfy $\mathbb{P}\{S_{\text{stap,post}} \geq \tau \mid H_0\} = \alpha$, changes in TPR reflect background-tail compression despite scores never increasing.

3.3 Algorithmic integrity and deterministic safeguards

To make the architecture auditable and to avoid leakage or discretization artifacts masquerading as performance gains, we enforce deterministic, label-free safeguards:

- **Conditional execution as a decision-rule variant.** To reduce latency, STAP can be skipped outside a flow-proxy region and replaced by the baseline score there. Because this changes the overall decision rule, any conditional-execution reporting is treated as a distinct algorithmic variant and is accompanied by leakage controls (same-window vs. disjoint-window vs. randomized proxy masks). Unless otherwise stated, conditional execution uses *disjoint-window* proxy gating for detection reporting.
- **Short-ensemble band discretization sanity.** For extremely short ensembles (e.g. $N \in \{5, 9, 17\}$), the coarse frequency resolution $\Delta f = \text{PRF}/N$ can cause requested bands to collapse or overlap after discretization. If the discretized bands are empty, DC-only, or overlapping, KA is disabled ($w \equiv 1$) with an explicit reason label.
- **Proxy-mask circularity control.** Proxy maps are never used to compute ROC/TPR/FPR; they are used only to define compute masks (conditional execution) or protected/candidate sets (KA), which are audited separately. Quantitative ROC claims are restricted to regimes with independent H_1/H_0 labels (simulator truth or B-mode structural geometry).

For auditability, we log explicit reason labels so that "KA off" ($w \equiv 1$) is an expected success mode under flat or non-actionable evidence. Supplementary Material provides modality-agnostic sufficient conditions and additional sentinels used to interpret when the KA prior should be active or inert.

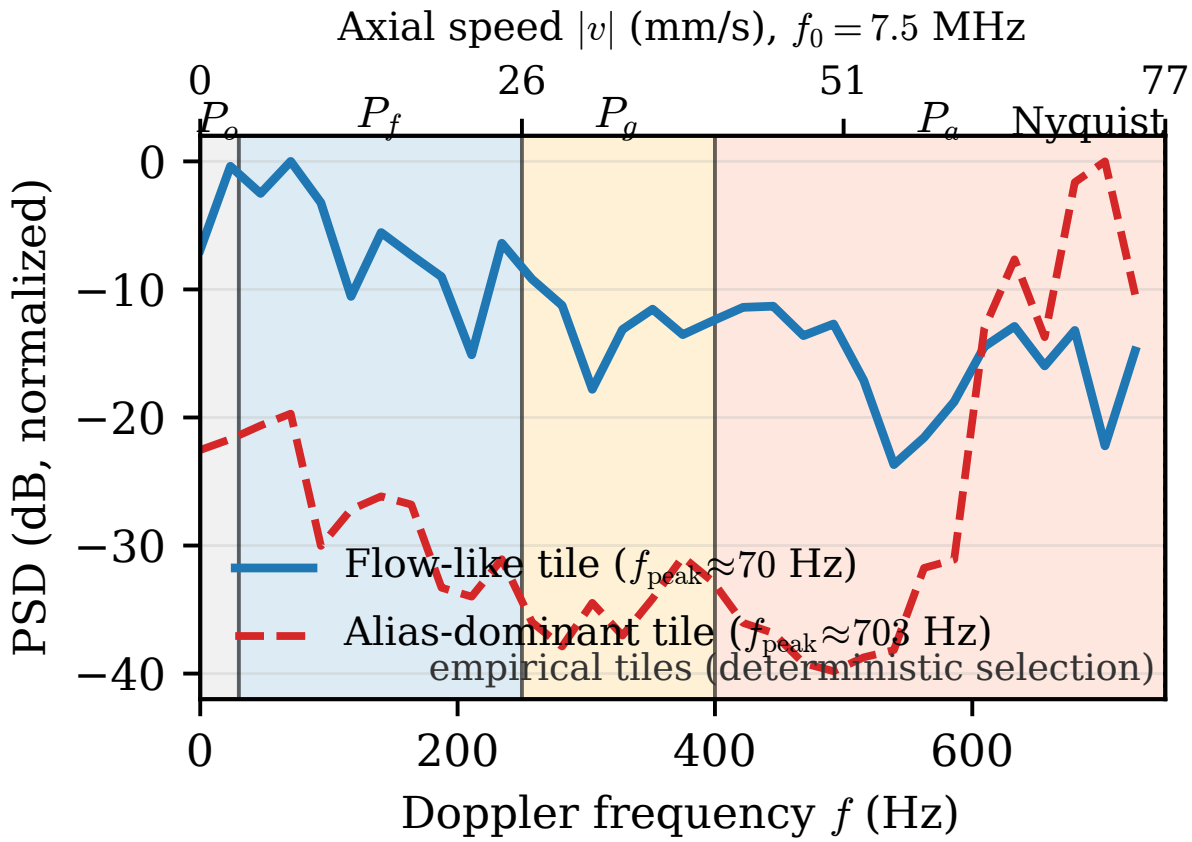

Figure 3: Doppler power spectral density (PSD) partitioning and kinematic band geometry (empirical example). Curves show one-sided PSDs of the tile-mean slow-time signal on a BrainAliasContract k-Wave pilot window (PRF = 1500 Hz): a flow-like tile (peak in P_f) and an alias-dominant tile (peak in P_a). The Nyquist interval is deterministically partitioned into DC/high-amplitude clutter (P_o), microvascular flow (P_f), a transition guard band (P_g), and aliased high-velocity artifacts or vibration (P_a). Auxiliary band-energy descriptors (e.g. alias-to-flow ratios) are used for interpretation and, optionally, for shrink-only tail regularization (Supplementary Material).

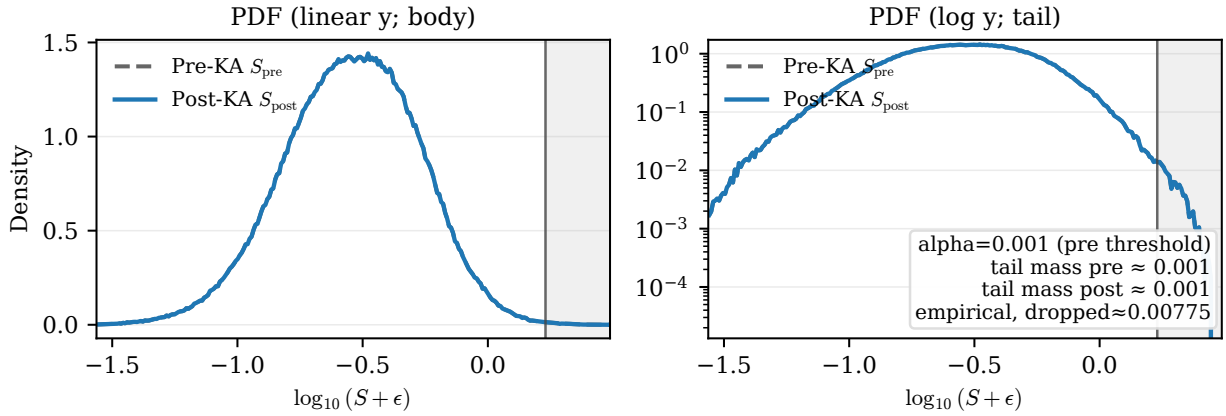

Figure 4: Empirical statistical effect of the strictly bounded, shrink-only penalty on a heavy-tailed background-proxy score distribution (Twinkling calculi artifact sequence). The pre-KA matched-subspace score $S_{\text{stap,pre}}$ exhibits a heavy right tail characteristic of structured artifacts. Applying the non-increasing penalty $\gamma(i) \geq 1$ yields the post-regularization score $S_{\text{stap,post}} = S_{\text{stap,pre}}/\gamma(i)$, compressing the background tail while preserving an explicit do-not-penalize protected set (scores unchanged on high-confidence flow locations). The vertical line indicates the pre-KA threshold τ_α chosen so that $\mathbb{P}\{S_{\text{stap,pre}} \geq \tau_\alpha \mid \text{bg proxy}\} = \alpha$ (here $\alpha = 10^{-3}$); the shaded region highlights the corresponding right tail, and the post-KA tail mass is evaluated at the same τ_α.

4 Experimental Design

4.1 Fixed-profile protocol and operating-point reporting

All hyperparameters (tile geometry, aperture, band definitions, conditioning targets, and prior thresholds) are fixed per regime before reporting results. Detection is evaluated by thresholding the high-score tail (right tail) at fixed FPR targets:

$$\mathbb{P}\{S \geq \tau_\alpha \mid H_0\} = \alpha, \qquad \mathrm{TPR}(\alpha) = \mathbb{P}\{S \geq \tau_\alpha \mid H_1\}.$$

For each experiment, we report the supported floor $fpr_{\min} = 1/n_{\mathrm{neg}}$ and bootstrap uncertainty at the natural unit of variation (windows or frames, depending on regime). These choices enforce a strict low-FPR operating discipline and avoid over-interpreting unsupported tail points.

Threshold calibration and deployment. Unless otherwise stated, τ_α is calibrated per window (or per cine sequence) on that unit's negative set to report $\mathrm{TPR}(\alpha)$ at the intended operating point; in labeled regimes, the negative set is available for *exact* calibration, while in deployment it would be estimated via a CFAR-style training procedure or from a fixed calibration set. This per-window protocol should therefore not be read as requiring oracle per-window tuning in deployment. We accordingly report a cross-window threshold-transfer audit for Brain-* pilots (Table 4), which calibrates τ_α on one window and evaluates on a disjoint window and quantifies the resulting realized FPR drift.

4.2 Integrity checks and deterministic sentinels

This work is a reproducible processing pipeline (decode → baseline → STAP score → optional KA regularization). Several practical pathologies can invalidate tail-rate reporting or spuriously activate the KA layer (e.g. missing acquisition metadata, coarse short-ensemble band discretization, proxy-mask circularity, or compute-gating leakage). We therefore enforce deterministic, label-free integrity checks that either (i) record provenance and proceed conservatively or (ii) disable KA ($w \equiv 1$) with an explicit reason label. A concise checklist (and where each sentinel is logged) is provided in Supplementary Material.

4.3 Regimes and claim scope

Claims are stratified by regime to separate quantitative detection efficacy from label-free instrumentation.

- **Brain-k-Wave simulations (labeled).** Across Brain-OpenSkull / Brain-AliasContract / Brain-SkullOR regimes, the fixed profile retains measurable sensitivity at ultra-low FPR in the matched-subspace STAP score (i.e., $\mathrm{TPR}(\alpha) > 0$ as $\alpha \to 10^{-4}$–10^{-3}), with the strongest low-FPR gains in the alias-augmented Brain-AliasContract and skull/OR-inspired Brain-SkullOR regimes (Table 6).
- **Phantom acquisitions (structurally labeled Gammex flow phantom).** We decode raw beamformer outputs, derive B-mode-only tube masks (flow/background), benchmark conventional short-ensemble Doppler baselines, and report structural-label low-FPR operating points and within-ensemble motion ladders.

Regime	Data type	Claim scope
Brain-* (k-Wave)	simulated IQ	low-FPR ROC and latency (simulation)
Gammex flow phantom	real Doppler IQ (phantoms)	structural-label ROC + motion ladders (phantom; not clinical)
Accepted SIMUS-ClinUtility-v2	moving-scatterer SIMUS/PyMUST	held-out structural and functional checkpoint on accepted nuisance profiles
ULM Zenodo 7883227	real kHz IQ	motion robustness proxies (no task ROC)
Supplementary datasets	real IQ and PD-only	descriptive/audit analyses (no task ROC)

Table 2: Claims discipline by regime. Quantitative detection performance is reported on labeled Brain-* simulations, structurally labeled phantom data (Gammex flow phantom), and the accepted clinically anchored *SIMUS-ClinUtility-v2* moving-scatterer benchmark. ULM 7883227 is used for label-free motion-robustness evidence on real kHz IQ. Additional datasets are used only for descriptive/audit analyses and are reported in Supplementary Material.

Regime (sampling)	Baseline	STAP core (fixed profile)	Doppler bands (Hz)
Brain-* (k-Wave; PRF 1500; 64-frame windows)	MC–SVD (energy fraction $e = 0.90$); adaptive SVD (SV similarity cutoff); block-wise local SVD (overlap-add; $e = 0.90$); RPCA; HOSVD (baselines scored by PD on residual)	tiles 8×8, stride 3; $L_t = 8$; robust covariance (Tyler-type) with diagonal loading $\lambda = 0.07$; matched-subspace flow-band score	$P_f = [30, 250]$, guard $[250, 400]$, $P_a = [400, 750]$
Gammex flow phantom (linear17; PRF 2500; $N = 17$ shots; pooled over cine frames)	baseline SVD band-pass keep $[2, 17]$; conventional short-ensemble comparators (PD/log-PD/Kasai)	tiles 8×8; $L_t = 16$; Tyler-type covariance with $\lambda = 0.07$; stride 6 (along) / 4 (across); matched-subspace flow-band score	$P_f = [150, 450]$, guard $[450, 700]$, $P_a = [700, 1200]$
ULM Zenodo 7883227 (PRF 1000; 128-frame blocks)	MC–SVD baseline with energy fraction $e = 0.975$ frozen by a one-time label-free sweep	tiles 8×8, stride 3; $L_t = 64$; SCM covariance with $\lambda = 0.07$; matched-subspace flow-band score	$P_f = [10, 150]$, guard $[150, 200]$, $P_a = [200, 500]$

Table 3: Frozen operating profiles for the main regimes. All settings are fixed per regime prior to reporting results; exact commands and exported metadata are provided in Supplementary Material.

- **Accepted SIMUS-ClinUtility-v2 benchmarks (structurally/ functionally labeled moving-scatterer data).** We report held-out structural and functional detection performance only on the accepted nuisance profiles (`ClinMobile-Pf-v2`, `ClinIntraOpParenchyma-Pf-v3`), after re-freezing all method families on the accepted development split.
- **ULM Zenodo 7883227 (real kHz IQ).** We report label-free motion-robustness curves on beamformed IQ blocks using controlled rigid/nonrigid motion injection, with fairness alignment and degeneracy checks (no task ROC).
- **Supplementary analyses.** Additional label-free audits (e.g. Shin Fig3 real IQ; PD-only whole-brain fUS) and diagnostic panels are provided in Supplementary Material.

5 Results

Results roadmap. We first report labeled Brain-* simulation results at ultra-low FPR, then structurally labeled Gammex flow phantom results (including motion ladders), then the accepted clinically anchored *SIMUS-ClinUtility-v2* checkpoint, and finally label-free motion-robustness evidence on real kHz IQ (ULM 7883227). Additional dataset audits and diagnostic panels are provided in Supplementary Material.

Scope and low-FPR operating protocol. Quantitative detection performance is reported on labeled Brain-* simulations, structurally labeled phantom regimes (Gammex flow phantom sequences; labels derived from B-mode geometry), and the accepted *SIMUS-ClinUtility-v2* benchmark using the strict low-FPR operating protocol of Section 4.1. Real kHz IQ datasets are used for instrumentation (band-energy integrity checks and label-free motion-robustness proxies), not task ROC. Conditional execution is a decision-rule variant (compute gating) that uses a flow-proxy mask; leakage ablations and additional diagnostics are reported in Supplementary Material.

5.1 Brain-k-Wave regimes (labeled): low-FPR ROC and latency

We report labeled low-FPR operating points on Brain-* k-Wave simulations using flow-evidence detection scores under a right-tail convention (higher = more flow-like; thresholds calibrated on the high-score tail): a baseline power Doppler score and the proposed matched-subspace STAP score (Notation). Baseline methods compare clutter suppression strategies (MC–SVD; adaptive SVD with a spatial singular-vector similarity cutoff; block-wise local SVD with overlap-add; RPCA; HOSVD) and are scored using the same power Doppler statistic; STAP is evaluated on top of the MC–SVD residual. Across Brain-* regimes, these baselines (including adaptive and local SVD filters) degrade sharply in the ultra-low-FPR tail, while the localized STAP score retains measurable sensitivity at ultra-low FPR (i.e., $\mathrm{TPR}(\alpha) > 0$ at $\alpha = 10^{-4}$–10^{-3}), including in the SkullOR regime where alias deliberately leaks into the positive class. Table 6 summarizes low-FPR operating points for representative pilot realizations over five non-overlapping 64-frame windows (offsets 0/64/128/192/256); each window has $n_{\mathrm{neg}} = 50199$ background pixels ($fpr_{\min} = 1/n_{\mathrm{neg}} \approx 1.99 \times 10^{-5}$), so FPR targets down to 10^{-4} are empirically supported. Numbers are medians (IQR) over windows and are computed by choosing τ per method per window on that window's negative set to match the target FPR, then evaluating TPR on the corresponding positive set. The wider IQR at $\alpha = 10^{-4}$ in Brain-OpenSkull reflects window-to-window nonstationarity and finite tail support under a fixed, local-covariance profile (some windows with nonzero supported-tail sensitivity versus near-floor windows). Per-window thresholds and realized FPRs are provided in Supplementary Material. Importantly, the MC–SVD+PD baseline row and the STAP row share the identical MC–SVD residual cube and differ only in the detector statistic (PD versus matched-subspace), isolating the effect of the detector rather than the clutter-suppression stage. To quantify deployment-relevant threshold transfer, Table 4 calibrates τ_α on one window and evaluates on a disjoint window, reporting both cross-window TPR and the realized cross-window FPR (FPR drift). Full per-pair outputs are written to `reports/brain_crosswindow_calibration.csv`.

Baseline fairness and frozen settings. All baseline hyperparameters are fixed prior to reporting results. For Brain-* pilots, the MC–SVD energy-fraction cutoff is selected by a one-time sweep and then frozen at $e = 0.90$ for all reported windows (sweep report in `reports/brain_mcsvd_`

Regime	α	within-window TPR med (IQR)	cross-window TPR med (IQR)	cross-window realized FPR med (IQR)
Brain-OpenSkull	10^{-4}	0.0730 (0.0008,0.0865)	0.0621 (0.0000,0.1839)	1.20e-04 (0,3.29e-01)
Brain-OpenSkull	3×10^{-4}	0.1288 (0.0012,0.2222)	0.1409 (0.0000,0.3285)	3.49e-04 (0,3.32e-01)
Brain-OpenSkull	10^{-3}	0.2218 (0.0015,0.4368)	0.2785 (0.0000,0.4883)	1.06e-03 (0,3.34e-01)
Brain-AliasContract	10^{-4}	0.6098 (0.4790,0.6755)	0.6363 (0.4344,0.7441)	9.96e-05 (1.99e-05,6.72e-04)
Brain-AliasContract	3×10^{-4}	0.6867 (0.6090,0.7493)	0.7351 (0.5552,0.8146)	3.78e-04 (5.48e-05,9.06e-04)
Brain-AliasContract	10^{-3}	0.8301 (0.7847,0.8820)	0.8404 (0.7236,0.9022)	1.04e-03 (2.99e-04,2.49e-03)
Brain-SkullOR	10^{-4}	0.5871 (0.5006,0.6682)	0.5827 (0.4474,0.7139)	1.29e-04 (1.99e-05,4.48e-04)
Brain-SkullOR	3×10^{-4}	0.7205 (0.6048,0.7785)	0.6801 (0.5755,0.7937)	3.19e-04 (5.48e-05,6.92e-04)
Brain-SkullOR	10^{-3}	0.8731 (0.8189,0.8758)	0.8516 (0.7726,0.9056)	1.02e-03 (6.52e-04,1.40e-03)

Table 4: Brain-* cross-window threshold calibration audit for the pre-KA STAP score. "Within-window" matches the main reporting protocol (calibrate τ_α on a window's negatives and evaluate on that same window) and is identical to the STAP row of Table 6 (same score, masks, and five 64-frame windows). "Cross-window" calibrates τ_α on one window and evaluates on a disjoint window ($n = 20$ ordered window pairs). Reported values are median (IQR) over windows/pairs, and the realized cross-window FPR quantifies drift under window-to-window distribution shift.

`energy_sweep_seed1.csv` / `reports/brain_mcsvd_energy_sweep_seed1.json`). The adaptive similarity-cutoff SVD and block-wise local SVD baselines use deterministic, frozen rules; implementation details are provided in Supplementary Material and Supplementary Material. RPCA and HOSVD baselines use standard, reproducible settings documented in Supplementary Material. All baselines are scored using the same right-tail power Doppler statistic so that differences isolate the clutter-suppression stage rather than the scoring rule. Table 5 summarizes the baseline families and what differs between them.

Detector-component ablation (whitening versus band selectivity). To address the concern that the proposed score is "local adaptive whitening + band energy" rebranded, we report two detector ablations on the identical MC–SVD residual cube: (i) *whitened total power* (apply $R_i^{-1/2}$ but do not partition the Doppler spectrum) and (ii) an *unwhitened* matched-subspace ratio (set $R_i = I$ but retain the fixed flow-band subspace). Table 7 shows that whitening alone without band partition does not control the strict tail, while matched-subspace (banded) scoring is the dominant contributor in these Brain-* regimes under the fixed-profile protocol. The unwhitened ratio corresponds to a normalized Doppler-band energy detector; the full STAP score generalizes this statistic to colored interference via tile-local covariance whitening. In these pilot simulations, the strong performance of the identity-covariance control suggests that after coarse baseline filtering the residual can be close to temporally white; we retain whitening in the primary score as the statistically principled colored-interference generalization.

Detector-swap composability audit (matched-subspace scoring on competing residuals). To isolate detector effects from the clutter-suppression stage, we also compute the matched-subspace STAP score on residuals produced by competing clutter filters (RPCA and HOSVD)

Method	scope	cutoff / estimator	tuned?	score	specified
MC–SVD	global	fixed energy fraction ($e = 0.90$)	tune-once	PD	Supplementary Material
Adaptive SVD (SV similarity cutoff)	global	rank via spatial singular-vector similarity drop	deterministic	PD	Supplementary Material
Block-wise local SVD (overlap-add)	local	tile-local energy fraction ($e = 0.90$) + overlap-add	deterministic	PD	Supplementary Material
RPCA	global	PCP (IALM) with fixed iterations/tolerance	deterministic	PD (sparse)	Supplementary Material
HOSVD	global	tensor energy fractions (default $0.99, 0.99, 0.99$)	deterministic	PD (residual)	Supplementary Material
STAP (pre-KA)	local	robust covariance → whitening → flow-subspace ratio	fixed profile	matched-subspace	Section 3

Table 5: Summary of baseline comparators and detector statistics used in the Brain-* labeled regimes. Baseline rows share the same PD flow-evidence score on their respective residual cubes; STAP differs by applying tile-local covariance whitening and a matched-subspace (flow-subspace) score on top of the MC–SVD residual.

	Brain-OpenSkull			Brain-AliasContract			Brain-SkullOR		
Method / score	TPR@10^{-4}	TPR@3×10^{-4}	TPR@10^{-3}	TPR@10^{-4}	TPR@3×10^{-4}	TPR@10^{-3}	TPR@10^{-4}	TPR@3×10^{-4}	TPR@10^{-3}
MC–SVD	0.0000 (0.0000,0.0000)	0.0000 (0.0000,0.0000)	0.0000 (0.0000,0.0000)	0.0000 (0.0000,0.0000)	0.0000 (0.0000,0.0000)	0.0000 (0.0000,0.0000)	0.0000 (0.0000,0.0000)	0.0000 (0.0000,0.0000)	0.0000 (0.0000,0.0000)
Adaptive SVD (spatial singular-vector similarity cutoff)	0.0000 (0.0000,0.0000)	0.0000 (0.0000,0.0000)	0.0000 (0.0000,0.0000)	0.0000 (0.0000,0.0000)	0.0000 (0.0000,0.0000)	0.0000 (0.0000,0.0000)	0.0000 (0.0000,0.0000)	0.0000 (0.0000,0.0000)	0.0000 (0.0000,0.0000)
Block-wise local SVD (overlap-add)	0.0000 (0.0000,0.0000)	0.0000 (0.0000,0.0000)	0.0000 (0.0000,0.0000)	0.0000 (0.0000,0.0000)	0.0000 (0.0000,0.0000)	0.0000 (0.0000,0.0000)	0.0000 (0.0000,0.0000)	0.0000 (0.0000,0.0000)	0.0000 (0.0000,0.0000)
RPCA	0.0000 (0.0000,0.0046)	0.0015 (0.0000,0.0088)	0.0042 (0.0042,0.0173)	0.0000 (0.0000,0.0000)	0.0000 (0.0000,0.0000)	0.0000 (0.0000,0.0000)	0.0000 (0.0000,0.0000)	0.0000 (0.0000,0.0000)	0.0000 (0.0000,0.0000)
HOSVD	0.0000 (0.0000,0.0015)	0.0015 (0.0015,0.0015)	0.0046 (0.0046,0.0077)	0.0015 (0.0015,0.0015)	0.0015 (0.0015,0.0015)	0.0031 (0.0015,0.0046)	0.0000 (0.0000,0.0015)	0.0000 (0.0000,0.0015)	0.0015 (0.0000,0.0031)
STAP (pre-KA matched-subspace) on MC–SVD residual	0.0730 (0.0008,0.0865)	0.1288 (0.0012,0.2222)	0.2218 (0.0015,0.4368)	0.6098 (0.4790,0.6755)	0.6867 (0.6090,0.7493)	0.8301 (0.7847,0.8820)	0.5871 (0.5006,0.6682)	0.7205 (0.6048,0.7785)	0.8731 (0.8189,0.8758)

Table 6: Brain-* k-Wave low-FPR operating points using flow-evidence detection scores (higher = more likely flow): baseline power Doppler and matched-subspace STAP. Each regime uses five non-overlapping 64-frame windows (offsets 0/64/128/192/256); each window has $n_{\text{neg}} = 50199$ ($fpr_{\min} \approx 1.99 \times 10^{-5}$). At this support, FPR targets 10^{-4}, 3×10^{-4}, and 10^{-3} correspond to ≈ 5, ≈ 15, and ≈ 50 false-positive pixels per window, respectively. Entries report median (IQR) TPR across windows at each target FPR. Baselines compare clutter suppression strategies (MC–SVD; adaptive SVD (SV similarity cutoff); block-wise local SVD (overlap-add); RPCA; HOSVD) and are scored by power Doppler; STAP is applied on top of MC–SVD. Full per-window outputs are provided in Supplementary Material.

and compare it to power Doppler scoring on the same residual under the identical window/mask protocol. Table 8 reports this paired-residual audit; under the fixed STAP profile used throughout, strict-tail improvements do not automatically persist when composing STAP with these more aggressive low-rank baselines. We therefore treat STAP as a second-stage detector defined for a coarse baseline residual (MC–SVD in the main pipeline), and use the controlled detector swap in Table 6 (PD versus matched-subspace on the same MC–SVD residual) as the primary isolation of detector effects.

Covariance-training contamination ablation (Brain-*). Because the detector relies on tile-local covariance estimation, a potential concern is "self-training" or training-set contamination: if high-flow snapshots dominate $\widehat{R}_i$, whitening could suppress the very structure being detected or yield brittle thresholds. In the main profile we estimate R_i from all Hankelized snapshots within each tile (no oracle exclusion). As a guard-cell-style ablation, we re-estimate the robust covariance after trimming the top $q = 0.05$ highest-energy training snapshots (per tile) and leave the detector statistic unchanged. Table 9 reports the resulting strict-tail operating points; in these Brain-* pilots the trim ablation does not qualitatively change the supported low-FPR behavior, suggesting that the reported strict-tail gains are not an artifact of covariance self-training. We further report

Missing table `reports/brain_detector_ablation_table.tex`.
Reproduce it with `PYTHONPATH=.pythonscripts/brain_detector_ablation_table.py`.

Table 7: Detector-component ablation on the identical MC–SVD residual cube: whitened total power, unwhitened matched-subspace ratio, and full matched-subspace STAP.

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Reproduce it with `PYTHONPATH=.pythonscripts/brain_detector_swap_table.py`.

Table 8: Paired-residual detector swap on competing clutter-filter outputs, comparing power Doppler and matched-subspace STAP on the same residual cube.

a 1D fixed-profile sensitivity sweep over covariance diagonal loading λ, tile geometry, and slow-time aperture L_t on the canonical Brain-OpenSkull pilot (Supplementary Material).

Relaxed operating-point sanity anchor (Brain-*). Several baseline rows in Table 6 take the value $\mathrm{TPR}(\alpha) \approx 0$ at $\alpha \leq 10^{-3}$. To reduce ambiguity around whether these zeros reflect implementation or tuning errors, Table 10 contrasts one strict-tail operating point ($\alpha = 10^{-3}$) with two relaxed operating points ($\alpha \in \{10^{-2}, 10^{-1}\}$) under the same window protocol; $\alpha = 10^{-2}$ is a moderate reference point and $\alpha = 10^{-1}$ is an intentionally relaxed non-degeneracy sanity anchor.

Missing table `reports/brain_baseline_sanity_relaxed_table.tex`.
Reproduce it with `PYTHONPATH=.pythonscripts/brain_baseline_sanity_table.py`.

Why PD-after-filter baselines can collapse in the strict tail. A baseline clutter filter can yield nontrivial separation at moderate operating points yet still fail at ultra-low FPR if the background score distribution has an extreme right tail: the background-calibrated threshold τ_α (e.g. the 99.9th percentile of H_0 at $\alpha = 10^{-3}$) can lie above essentially all H_1 pixels, yielding $\mathrm{TPR}(\alpha) \approx 0$ by construction. For example, in a representative Brain-OpenSkull window the MC–SVD+PD baseline background quantiles are $q_{0.99} \approx 5.9$ and $q_{0.999} \approx 9.3$ while the flow $q_{0.999} \approx 0.094$, so no flow pixels exceed the $\alpha = 10^{-3}$ threshold; Figure 5 visualizes the corresponding tail geometry ($\tau_{10^{-3}} > \max(H_1)$) for the baseline PD score and contrasts it with the STAP score computed on the same residual. This tail-dominance failure mode motivates both the strict low-FPR operating discipline and detectors that explicitly reshape the tail (here: tile-local covariance whitening and matched-subspace scoring).

Latency and conditional execution. We also report representative runtimes for baseline clutter suppression and STAP under a conservative fixed operating profile (tile geometry, slow-time aperture L_t, and covariance-conditioning targets fixed *a priori*). Conditional execution reduces runtime by skipping the STAP stage on tiles with negligible proxy flow support and reverting to the baseline score there. Because this changes the overall decision rule, it can change operating characteristics; we therefore treat conditional execution as a distinct variant and require explicit leakage controls when reporting it. Unless otherwise stated, conditional execution uses a *disjoint-window* proxy mask for detection reporting (mask derived from one window and applied to a disjoint window) to avoid same-window proxy leakage.

Missing table `reports/brain_cov_train_ablation_table.tex`.
Reproduce it by generating trimmed-STAP replay bundles with `PYTHONPATH=.condarun-nstap-fuspythonscripts/fair_filter_comparison.py--modematrix--eval-scorevnext--methodsstap_full,stap_covtrim_q05--generated-rootruns/pilot/fair_filter_matrix_pd_r3_localbaselines--autogen-missing--stap-devicecuda--out-csvreports/fair_matrix_covtrim.csv--out-jsonreports/fair_matrix_covtrim.json` and then running `PYTHONPATH=.pythonscripts/brain_cov_train_ablation_table.py`.

Table 9: Covariance-training contamination ablation on the Brain-* pilots, comparing the default covariance estimate with a trimmed high-energy training variant.

Latency is measured via deterministic replay scripts that run both a *legacy* path (eager execution; CUDA graphs disabled) and an *optimized* path (CUDA graphs enabled when supported; `STAP_FAST_CUDA_GRAPH=1`). We report *steady-state* means over windows/frames 2..N and exclude the first replay window/frame, which includes one-time GPU overheads (initialization, kernel compilation, and optional CUDA-graph capture). The baseline and STAP stages are timed separately (excluding disk I/O), and "Total" reports baseline+STAP wall time; full legacy-versus-optimized per-window statistics and parity checks for the RTX 4080 SUPER reruns are written to `reports/latency/4080super_latency_summary.csv` / `reports/latency/4080super_latency_summary.json`.

Scenario	Baseline (s)	STAP full (s)	Total full (s)	STAP conditional (s)	Total conditional (s)
Brain-OpenSkull (k-Wave; 64-frame window)	0.718	0.513	1.231	0.264	0.982
Brain-AliasContract (k-Wave; 64-frame window)	0.634	0.555	1.189	0.333	0.967
Brain-SkullOR (k-Wave; 64-frame window)	0.617	0.553	1.170	0.314	0.931
Shin RatBrain Fig3 (PRF 1000; 128-frame window; $L_t = 64$)	0.051	0.939	0.990	0.440	0.491
Gammex along (linear17 @ PRF 2500; per cine frame)	0.012	0.246	0.258	—	—
Gammex across (linear17 @ PRF 2500; per cine frame)	0.005	0.105	0.110	—	—

Table 11: Consolidated CUDA-accelerated latency summary (steady-state timing; excludes disk I/O). Times are steady-state means over windows/frames 2..N from a single replay process: Brain-* uses windows 2..5 (offsets 0/64/128/192/256) via `scripts/latency_rerun_check.py`, while Shin and Gammex use windows/frames 2..N via `scripts/latency_realdata_rerun_check.py`. The first replay window/frame includes one-time GPU overheads (initialization, kernel compilation, and optional CUDA-graph capture) and is excluded. "Total" reports baseline+STAP wall time. Hardware: NVIDIA GeForce RTX 3050 6GB Laptop GPU (CUDA 12.1; `stap_device=cuda`). Conditional execution is not reported for structurally labeled Gammex bundles.

Portable-workstation feasibility (tempered). A common critique of covariance-adaptive STAP is that local covariance conditioning and the associated matrix factorizations/inversions are prohibitively expensive when applied densely across an image grid. Tables 11 and 12 show that, with a batched GPU implementation and steady-state reporting, the Brain-* and Gammex

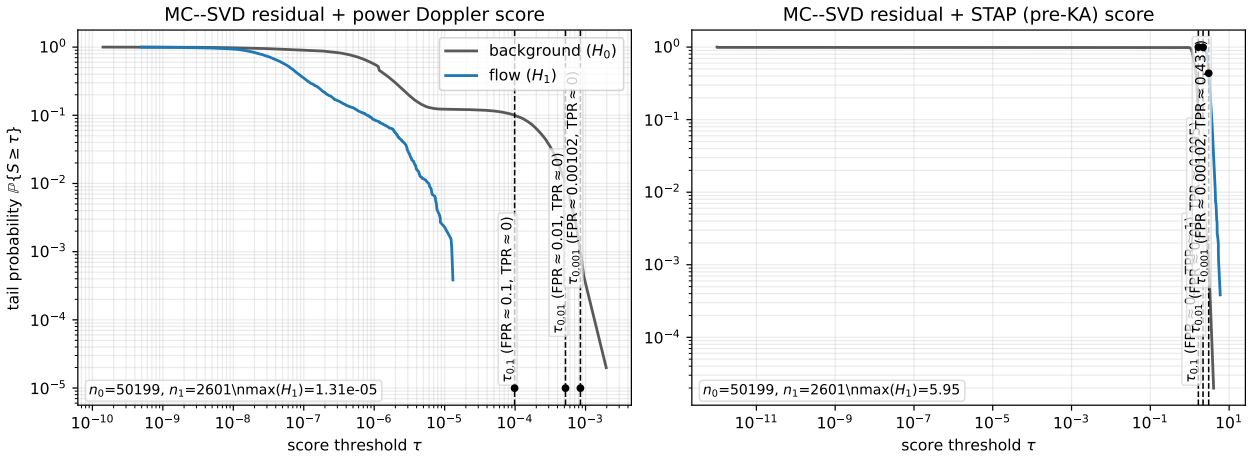

Figure 5: Representative Brain-OpenSkull window illustrating strict-tail collapse of PD-after-filter baselines. Curves show empirical tail probabilities $\mathbb{P}\{S \geq \tau\}$ for background (H_0) and flow (H_1) score distributions versus score threshold τ, for (left) the baseline power Doppler score on the MC–SVD residual and (right) the pre-KA matched-subspace STAP score computed on the same residual. Vertical lines mark background-calibrated thresholds τ_α at $\alpha \in \{10^{-1}, 10^{-2}, 10^{-3}\}$ and markers show the corresponding $\mathrm{TPR}(\alpha) = \mathbb{P}\{S \geq \tau_\alpha \mid H_1\}$. For the baseline PD score, $\tau_{10^{-3}}$ lies above $\max(H_1)$, yielding TPR ≈ 0 by construction; the STAP score retains nonzero sensitivity at the same strict-tail operating point.

regimes run in sub-second baseline+STAP wall time on both tested GPUs (Brain-* full totals ≈ 0.77–0.78 s on RTX 4080 SUPER), while Shin runs in ≈ 0.17 s total on RTX 4080 SUPER under the same steady-state replay protocol.[1] These results suggest that tile-local covariance adaptation need not require server-class hardware, but they also highlight that throughput remains strongly dataset/profile dependent and should be interpreted with the explicit decision-rule caveats of conditional execution.

A leakage ablation confirms that conditional execution can inflate apparent performance under same-window proxy gating, while leakage-controlled gating yields a small TPR decrease (order 10^{-3}) at the target FPRs; we therefore treat conditional execution as a distinct decision rule rather than a cost-free speedup. Table 13 summarizes the effect on Brain-OpenSkull pilots; full bootstrap CIs and details are provided in Supplementary Material.

[1]Latency is reported from steady-state replay (windows/frames 2..N) and excludes one-time GPU overheads (initialization, compilation, optional CUDA-graph capture) as well as disk I/O. For context, a 64-frame Doppler window at PRF 1500 Hz corresponds to $64/1500 \approx 43$ ms of acquired slow-time data, and a 128-frame window at PRF 1000 Hz corresponds to 128 ms. The present Python prototype therefore does not yet achieve real-time throughput at the full acquisition PRF, but it establishes practical feasibility on commodity laptop/workstation GPUs and motivates further optimization and tighter scanner integration.

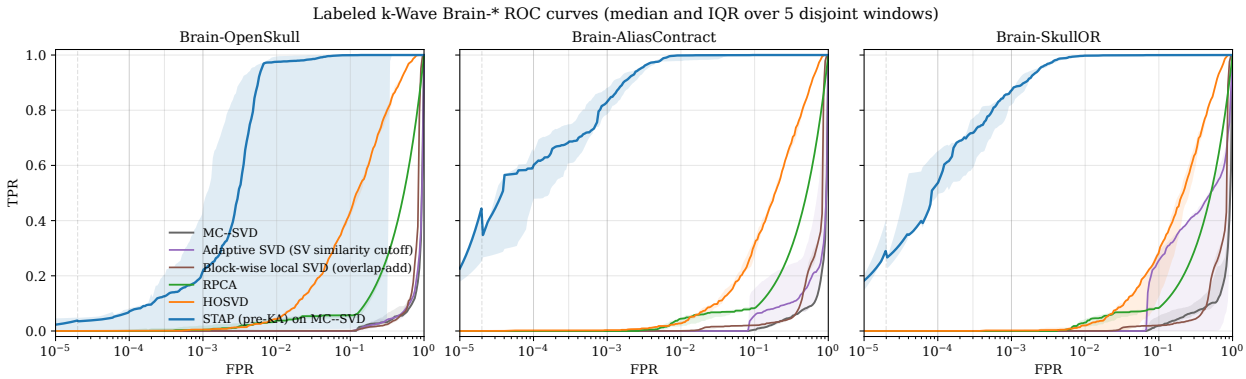

Figure 6: Brain-* k-Wave labeled regimes: ROC-style curves (TPR vs FPR on a log-x axis) under the same 64-frame window protocol used for the Brain-* low-FPR operating-point reporting in the main text. Curves include MC–SVD, adaptive and local SVD baselines, RPCA, HOSVD, and STAP; each curve shows the median and interquartile range over five disjoint windows. The moderate-FPR region illustrates that several PD-after-filter baselines can show separation at relaxed operating points while saturating at strict-tail operating points. The vertical dashed line indicates the supported floor $fpr_{\min} = 1/n_{\text{neg}}$.

Scenario	Baseline (s)	STAP full (s)	Total full (s)	STAP conditional (s)	Total conditional (s)
Brain-OpenSkull (k-Wave; 64-frame window)	0.230	0.538	0.768	0.334	0.556
Brain-AliasContract (k-Wave; 64-frame window)	0.235	0.544	0.779	0.372	0.607
Brain-SkullOR (k-Wave; 64-frame window)	0.242	0.542	0.783	0.374	0.607
Shin RatBrain Fig3 (PRF 1000; 128-frame window; $L_t = 64$)	0.020	0.154	0.174	0.116	0.136
Gammex along (linear17 @ PRF 2500; per cine frame)	0.007	0.053	0.060	—	—
Gammex across (linear17 @ PRF 2500; per cine frame)	0.001	0.019	0.020	—	—

Table 12: Same latency protocol as Table 11, rerun on a workstation-class GPU. Entries report optimized-path steady-state means (windows/frames 2..N; excludes disk I/O and excludes the first replay window/frame). Values are derived from `reports/latency/4080super_latency_summary.csv`. Hardware: NVIDIA GeForce RTX 4080 SUPER 16GB (see `reports/hw/4080super_nvidia_smi.txt`; `stap_device=cuda`). Conditional execution is not reported for structurally labeled Gammex bundles.

Conditional gating mask	ΔTPR@10^{-4} ($\times 10^{-3}$)	ΔTPR@3×10^{-4} ($\times 10^{-3}$)	ΔTPR@10^{-3} ($\times 10^{-3}$)
Same-window proxy mask	+0.31	+0.81	+3.05
Disjoint-window proxy mask	−3.05	−2.90	−1.74
Random mask (matched coverage)	−3.05	−2.97	−2.36

Table 13: Conditional-execution leakage ablation on Brain-OpenSkull pilots (conditional minus full STAP; five independent realizations). Entries report mean ΔTPR at fixed FPR targets. Same-window proxy gating can inflate apparent performance; disjoint-window proxy gating is used as the default leakage-controlled condition for robustness reporting. Full bootstrap CIs and details are provided in Supplementary Material.

5.2 Phantom acquisitions (Gammex flow phantom): structural ROC and within-ensemble motion ladders

Dataset and acquisition parameters. We use the open-access Doppler Twinkling Artifact database [40], which provides complex beamformer output from phantom acquisitions on a clinical scanner. Each cine frame contains a contemporaneous B-mode image and a short-ensemble color-flow (CFM) slow-time block with $N \in \{5, 9, 17\}$ pulses. Because Doppler PRF is not reliably available in the provided metadata for these acquisitions, all Twinkling experiments treat PRF as an explicit experimental assumption frozen per sequence family (listed in Supplementary Material) and enforce band-discretization sanity checks that disable KA ($w \equiv 1$) when discretized Pf/Pa bands are empty or overlapping (Section 3). Decoding/QC details are provided in Supplementary Material (PRF overrides and decoder sanity); reproduction commands and manifests are provided in Supplementary Material.

B-mode-only structural tube masks (independent H_1/H_0). For the Gammex flow phantom sequences we derive evaluation masks from B-mode structure only (no Doppler/STAP outputs). Using acquisition geometry metadata that specifies the CFM region-of-interest (ROI) within the B-mode grid (beam and sample offsets and any beam subsampling), we map the B-mode ROI onto the CFM grid and segment the tube lumen on a median log-envelope reference image. This yields (i) a conservative lumen (flow) mask, eroded to avoid wall pixels and (ii) a background mask that excludes a guard band around the lumen and bright reflectors. Mask provenance and QC metrics are recorded with each experiment; Figure 7 shows representative overlays. These structural labels support a defensible phantom ROC.

Frozen short-ensemble Doppler profile and baseline fairness scores. For the linear-probe Gammex sequences (PRF = 2500 Hz, $N = 17$ shots), we freeze a short-ensemble operating profile: baseline SVD band-pass keep range $[2, 17]$, STAP aperture $L_t = 16$, tiles 8×8, a Tyler-type robust covariance estimator, and diagonal loading $\lambda = 0.07$. To keep the profile auditable while avoiding sample-support limitations on small ROIs and excessive overlap/mixing on large ROIs, we use a deterministic geometry-only stride policy: choose the largest tile stride ≤ 6 such that the implied tile count is ≥ 500. On the two Gammex views this yields stride 6 (along; large ROI) and stride 4 (across; small ROI). Band definitions are frozen to $P_f = [150, 450]$ Hz and $P_a = [700, 1200]$ Hz. For baseline context on short ensembles, we also report conventional comparators derived from the baseline-filtered cube (power Doppler, log power Doppler, and Kasai lag-1 power). Reproducibility details are provided in Supplementary Material.

Structural-label low-FPR ROC (Gammex). We pool pixel scores across cine frames and report low-FPR operating points using right-tail thresholding on pooled background pixels, then evaluate TPR on pooled lumen pixels. In these short-ensemble Doppler acquisitions, the background set includes strong twinkling/artifact pixels whose right-tail scores dominate conventional baselines; consequently, low-FPR thresholds at $\alpha \leq 10^{-3}$ can lie above essentially all lumen pixels, yielding numerically zero baseline TPR at the reported operating points. Figure 8 visualizes this effect for a representative baseline comparator. To avoid treating pooled pixels as independent samples, we report 95% confidence intervals via a frame-level bootstrap: we resample cine frames as the unit of variation while holding the pooled threshold fixed. Table 14 summarizes the resulting structural-label operating points for the two Gammex views under their frozen profiles; full pooled

	Along (linear17 @ PRF 2500)			Across (linear17 @ PRF 2500)		
Method / score	TPR@10^{-4}	TPR@3×10^{-4}	TPR@10^{-3}	TPR@10^{-4}	TPR@3×10^{-4}	TPR@10^{-3}
Baseline (power Doppler)	0.000 [0.000,0.000]	0.000 [0.000,0.000]	0.000 [0.000,0.000]	0.000 [0.000,0.000]	0.000 [0.000,0.000]	0.000 [0.000,0.000]
Baseline (log power Doppler)	0.000 [0.000,0.000]	0.000 [0.000,0.000]	0.000 [0.000,0.000]	0.000 [0.000,0.000]	0.000 [0.000,0.000]	0.000 [0.000,0.000]
Baseline (Kasai lag-1 power)	0.000 [0.000,0.000]	0.000 [0.000,0.000]	0.000 [0.000,0.000]	0.000 [0.000,0.000]	0.000 [0.000,0.000]	0.000 [0.000,0.000]
STAP (pre-KA matched-subspace)	0.634 [0.628,0.640]	0.803 [0.798,0.808]	0.946 [0.943,0.948]	0.846 [0.842,0.850]	0.891 [0.888,0.893]	0.938 [0.936,0.940]

Table 14: Gammex flow phantom structural-label operating points using B-mode-derived structural masks. All scores are right-tail (higher indicates more flow evidence). Thresholds are chosen per method on pooled background pixels to match each target FPR and evaluated on pooled lumen pixels. Brackets show 95% frame-bootstrap CIs ($n = 2000$ resamples over cine frames) at the fixed pooled threshold. Pooled background counts are $n_{\mathrm{bg}} = 3.85 \times 10^6$ for the along view ($fpr_{\min} \approx 2.6\times10^{-7}$) and $n_{\mathrm{bg}} = 1.97\times10^6$ for the across view ($fpr_{\min} \approx 5.1\times10^{-7}$). Reproducibility details are provided in Supplementary Material.

summaries (including baseline-score variants and realized thresholds) are provided in Supplementary Material. Because STAP is evaluated on the same baseline-filtered residual cube, the strong STAP TPR at strict FPR rules out the interpretation that the baseline filter simply removed lumen flow; rather, the failure mode is heavy-tailed background scores dominating the right-tail threshold (see Figure 8).

Within-ensemble motion ladders (labeled, real acquisition). To obtain a labeled motion-robustness curve on a real acquisition without external motion ground truth, we inject synthetic *within-ensemble* motion directly into the CFM cube: for each cine frame we apply a per-shot rigid translation $(\Delta y_t, \Delta x_t)$ to the N CFM shots prior to baseline and STAP, with motion kinds `rw` (random walk) and `step`. Because the structural tube masks are derived from B-mode and fixed per frame, this produces a labeled "motion ladder" where the only change is within-ensemble corruption. We generate motion-corrupted ensembles (85 cine frames; amplitudes 0–2 px) and aggregate using frame-level bootstraps for CIs together with an explicit amp=0 equivalence check against the non-motion reference. Figure 11 shows the resulting curves; reproducibility details are provided in Supplementary Material.

Supplementary KA hygiene (artifact-heavy calculi). Additional KA hygiene analyses on artifact-heavy calculi sequences (region/reason summaries and representative pre/post tail examples) are provided in Supplementary Material.

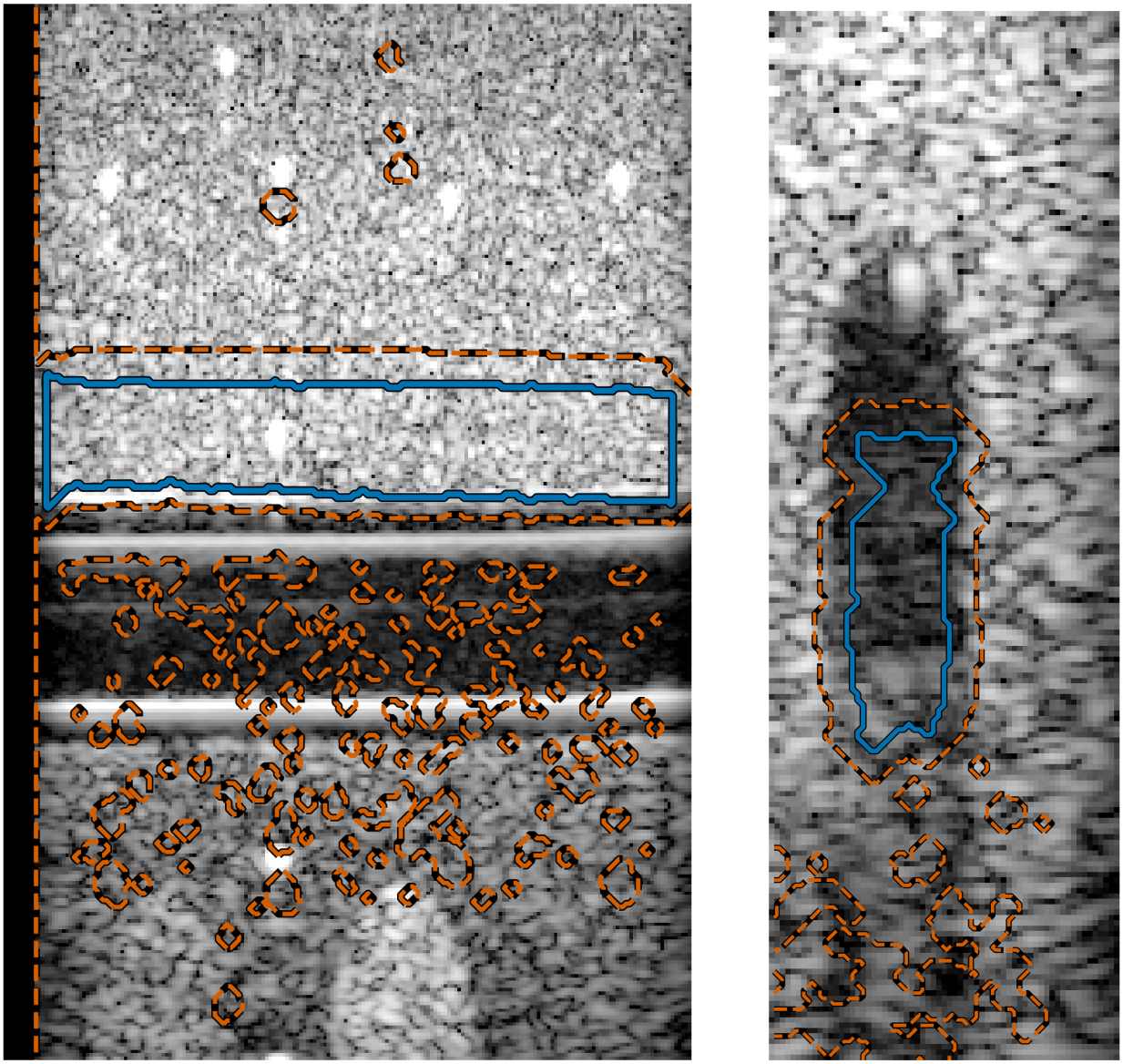

Figure 7: Gammex flow phantom structural evaluation masks derived from B-mode only. Left: along-linear view; right: across-linear view. The lumen mask is intentionally eroded to avoid wall/partial-volume pixels, and the background mask excludes a guard band around the lumen and bright reflectors. Images show the CFM-aligned B-mode ROI with contour overlays of the tube lumen (flow) mask (blue) and background mask (orange, dashed) used for structural-label ROC reporting. Reproducibility details are provided in Supplementary Material.

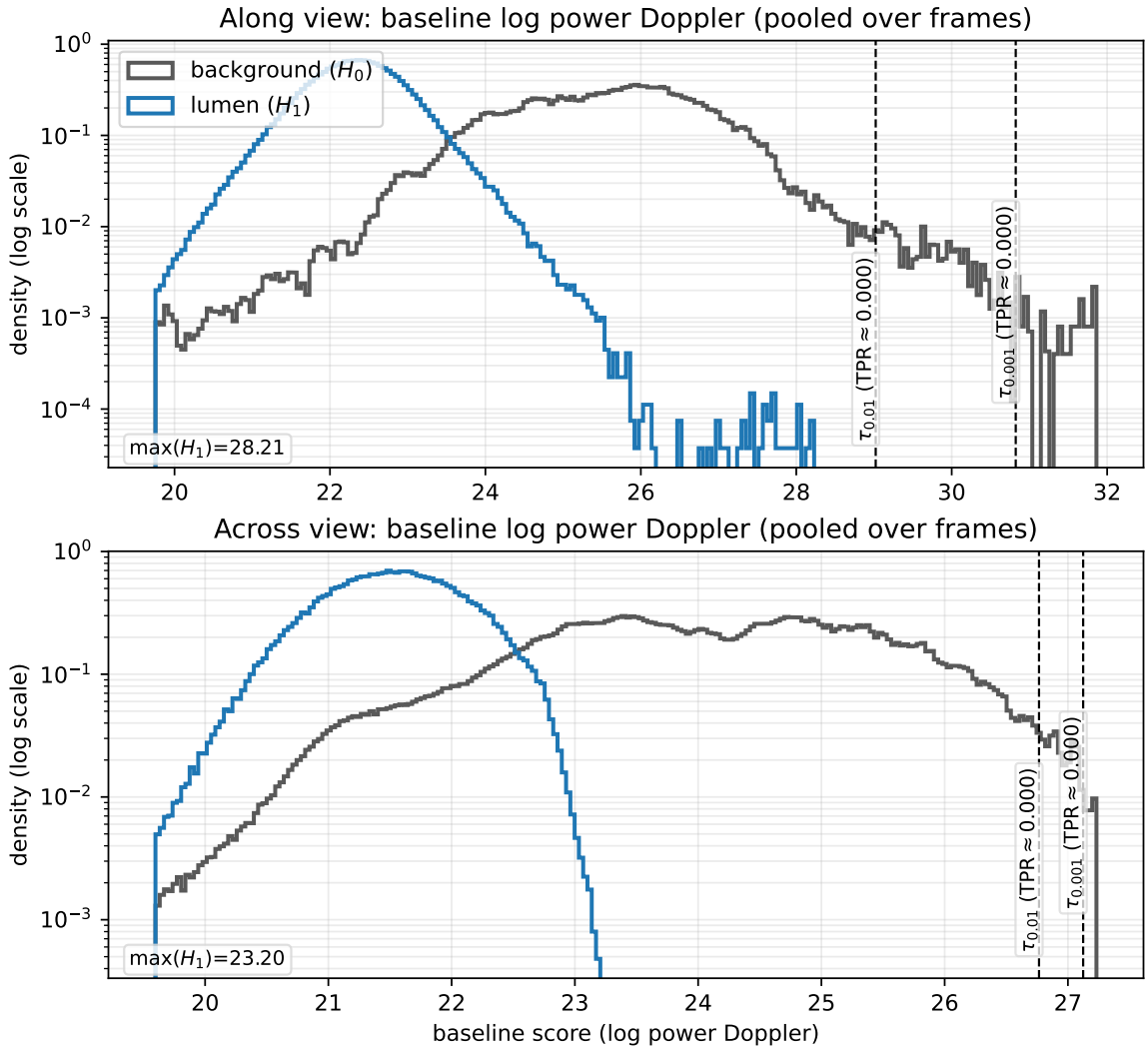

Figure 8: Why conventional baselines can have zero TPR at ultra-low FPR on Gammex short-ensemble data. The plot shows pooled score distributions for the baseline log power Doppler comparator on lumen (H_1; structural tube mask) versus background (H_0; structural background mask), for both along and across views. Vertical lines indicate background-calibrated right-tail thresholds τ_α at $\alpha \in \{10^{-2}, 10^{-3}\}$ (the main text reports $\alpha \leq 10^{-3}$; $\alpha = 10^{-2}$ is shown here for context). On both views the background tail threshold lies above $\max(H_1)$, so $\mathrm{TPR}(\alpha) = \mathbb{P}\{S \geq \tau_\alpha \mid H_1\}$ is numerically zero at these tail operating points.

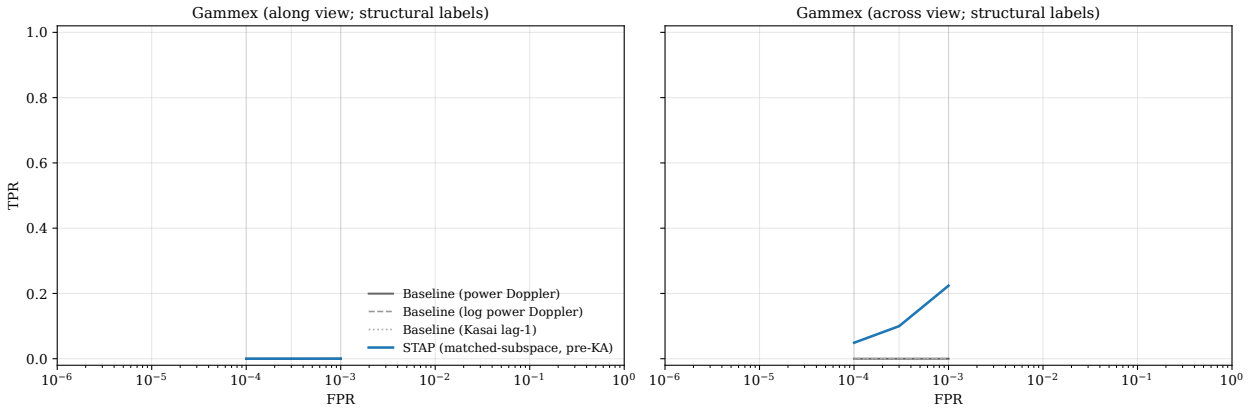

Figure 9: Gammex flow phantom structural labels: pooled ROC-style curves on along and across views, shown for baseline comparators and the pre-KA STAP matched-subspace score. The vertical guides indicate the strict low-FPR operating points reported for the structural-label evaluation in the main text.

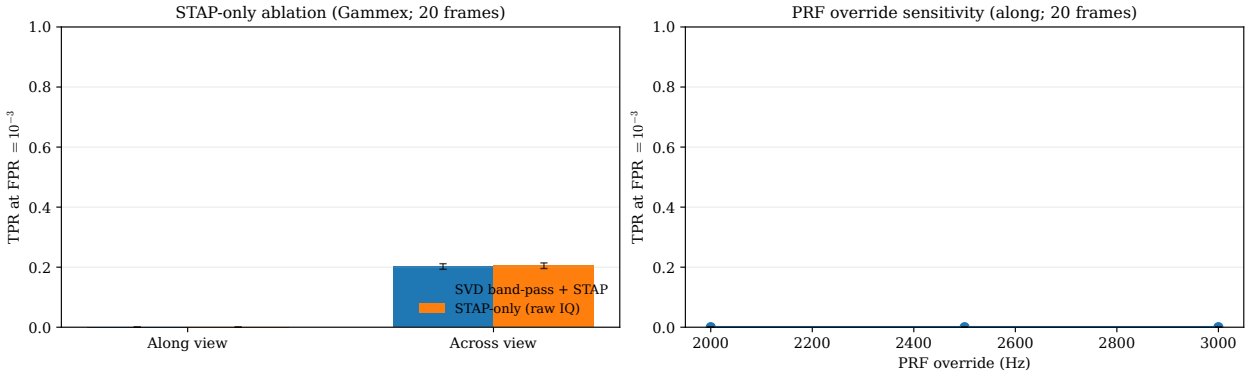

Figure 10: Gammex audits: STAP-only ablation and PRF override sensitivity (first 20 frames; along and across views). Left: STAP-only (raw IQ) vs baseline SVD band-pass + STAP at fixed FPR (10^{-3}). Right: PRF override sensitivity on the along view (PRF $\in \{2000, 2500, 3000\}$ Hz) under a frozen operating profile; this is a low-cost sanity audit of Doppler-band discretization, since PRF is constant within a sequence family and the discretization sentinels depend only on (N, PRF). Additional details are provided in Supplementary Material.

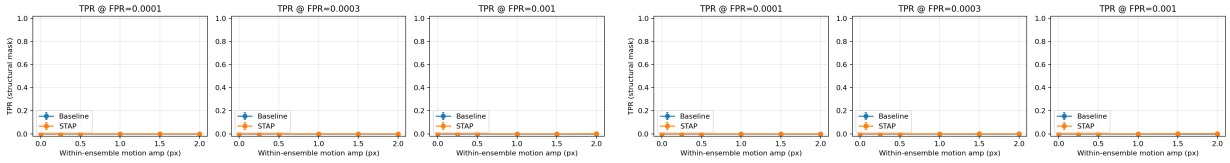

Figure 11: Gammex flow phantom within-ensemble motion ladder on the along-linear PRF=2500 sequence (85 frames). Left: random-walk per-shot translation; right: step translation. Each panel reports TPR vs motion amplitude at fixed FPR targets using B-mode structural masks. Shaded bands are 25–75% bootstrap intervals over cine frames. Reproducibility details are provided in Supplementary Material.

5.3 ULM Zenodo 7883227 (real kHz IQ): baseline calibration and nonrigid motion robustness

Dataset and loader. We use a real kHz IQ dataset released for ultrasound localization microscopy benchmarking [36, 37] and distributed as HDF5 blocks with accompanying acquisition metadata. Each block contains a dataset `iq` of shape $(2, H, W, T) = (2, 256, 180, 800)$ float32 representing I/Q over $T = 800$ slow-time frames. We load blocks on demand and return complex IQ as a cube of shape (T, H, W) (reproduction details in Supplementary Material).

One-time label-free baseline selection. For ULM we select a single MC–SVD energy-fraction cutoff by a one-time label-free sweep on blocks 1–3 over frames 0:128, and then freeze it for all subsequent ULM motion experiments. The sweep uses baseline-only runs and relies on label-free integrity and stability criteria (no task labels); it yields a frozen energy fraction $e = 0.975$ (full sweep details in Supplementary Material).

Nonrigid motion injection and fairness alignment ablation. We evaluate motion robustness by injecting controlled motion directly into the beamformed IQ cube and re-running baseline+STAP under a frozen ULM profile (PRF = 1000 Hz; $L_t = 64$; tiles 8×8; stride 3; SCM covariance; diagonal loading $\lambda = 0.07$; baseline MC–SVD with frozen $e = 0.975$). Because motion is injected at the beamformed IQ stage, this test stresses post-beamforming robustness to residual motion mismatch; it does not model channel-domain decorrelation or skull-induced phase-aberration physics. Motion ladders span two nonrigid families: `brainlike` (rigid random-walk translation plus a smooth, depth-decaying elastic warp with temporally correlated coefficients and micro-jitter) and `elastic` (nonrigid-only; rigid and micro-jitter disabled). For each amplitude we compute label-free robustness endpoints on the baseline score S_{base} and the matched-subspace STAP score $S_{\text{stap,pre}}$ relative to the no-motion reference under three *common* alignment modes (none / common B-mode / injected GT proxy) plus a diagnostic per-method "self" alignment. We additionally report degeneracy checks (std ratios, top-k overlap, and tail quantile ratios) to rule out "stability by collapse." Figure 12 shows the main `brainlike` sweep; additional motion variants are provided in Supplementary Material.

Endpoint definition. For a score map $S_{\text{meth}}(a)$ at motion amplitude a and the corresponding no-motion reference $S_{\text{meth}}(0)$ (same block), we compute $\rho_{\text{meth}}(a)$ as the Pearson correlation between the aligned maps on a central crop Ω_a that excludes a margin of $\lceil a \rceil + 2$ pixels to avoid boundary artifacts from warping. Correlations are computed directly on the exported right-tail scores (no log transform) and use the same crop/alignment for baseline and STAP in the "common" alignment modes.

Representative quantitative endpoints (ULM). At a 3 px motion amplitude under the injected GT alignment proxy, the median correlation on the baseline score S_{base} drops to ≈ 0.20 (`brainlike`), while the STAP score $S_{\text{stap,pre}}$ remains ≈ 0.75 (medians over blocks 1–3; full tabular summaries and additional motion variants are provided in Supplementary Material). Degeneracy checks confirm that this stability is not explained by map collapse: the median std ratio and 0.999-quantile ratio of $S_{\text{stap,pre}}$ remain nontrivial across amplitudes, and top-k overlap decays gradually rather than saturating to trivial values.

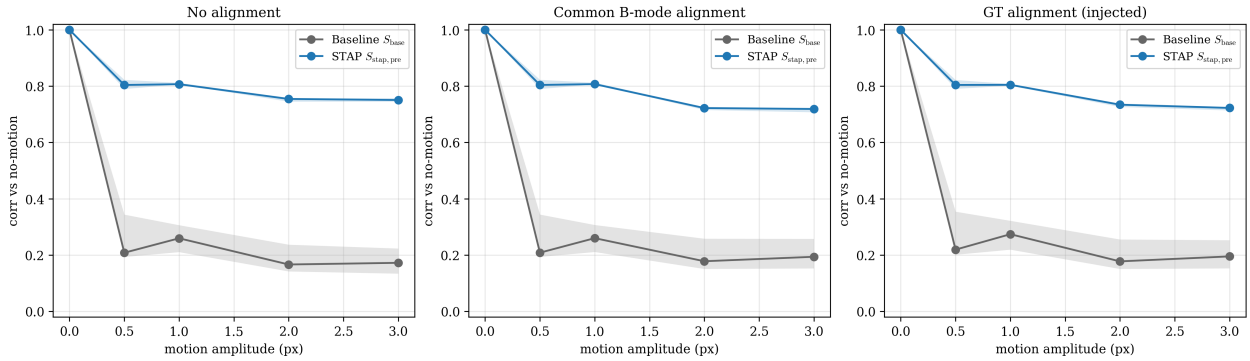

Figure 12: ULM 7883227 motion sweep (blocks 1–3; frames 0:128; frozen baseline $e = 0.975$): `brainlike` motion (rigid + elastic). Each panel reports correlation vs the no-motion reference under a different alignment mode (none / common B-mode / injected GT proxy) for the baseline score S_{base} and the STAP score $S_{\mathrm{stap,pre}}$; shaded bands show median±IQR over blocks. Reproducibility details are provided in Supplementary Material.

Supplementary datasets and diagnostics. Additional label-free audits (e.g. Shin Fig3 real IQ; PD-only whole-brain fUS) and diagnostic panels (e.g. conditional-execution leakage ablations, KA hygiene panels) are provided in Supplementary Material.

6 Discussion

A) Robustness in the ultra-low-FPR tail

This work adapts space–time adaptive processing (STAP), a foundational approach in radar target detection under heavy clutter, to functional ultrasound (fUS) slow-time flow detection. Under the strict low-FPR operating protocol (Section 4.1), we find that common global or semi-global clutter suppressors (MC–SVD, RPCA, HOSVD) can lose sensitivity in the extreme right tail, while the proposed tile-based matched-subspace STAP score retains measurable sensitivity at ultra-low FPR (i.e., nonzero TPR(α) at $\alpha \approx 10^{-4}$–10^{-3} under empirically supported tail floors). This behavior is demonstrated on labeled k-Wave simulations and structurally labeled *in vitro* phantoms, where flow/background masks are defined independently from Doppler/STAP outputs via simulator truth or B-mode geometry. The within-ensemble motion ladders on phantom acquisitions further indicate that STAP retains detection stability under controlled per-shot motion corruption, consistent with its reliance on localized covariance rather than global low-rank structure.

B) Knowledge-aided prior as a fail-safe

The optional knowledge-aided (KA) layer is designed as a conservative safety regularizer rather than an unconstrained signal amplifier. It acts as a shrink-only prior penalty with an auditable (region, reason) output: when spatial-spectral evidence is flat or non-actionable, the prior defaults to $\mathcal{R}_0$ (uninformative; $w \equiv 1$) and leaves the pre-KA detector score unchanged. When the prior lies in $\mathcal{R}_1/\mathcal{R}_2$, it preserves an explicit do-not-penalize protected set (scores unchanged on high-confidence flow locations) and applies only non-increasing shrink on high-risk regions, so that "KA does nothing" is an intended success mode out of regime. This posture prioritizes transparent regime diagnostics and avoids score inflation in regimes where spectral evidence cannot justify aggressive intervention.

C) Data limitations: missing raw high-PRF complex IQ

A key limitation in evaluating Doppler-side covariance detectors is field-wide data availability: most public *in vivo* fUS releases do not include raw, high-PRF complex IQ ensembles needed to evaluate covariance-based Doppler detectors with independent H_1/H_0 labels. Many releases are temporally downsampled, post-filtered, or provided only as power-Doppler equivalents, which can irreversibly destroy the high-rate complex phase history needed for spatiotemporal covariance estimation and band-energy integrity checks. Accordingly, while we report real-data instrumentation (band-energy integrity, prior-regime behavior, and label-free stability under controlled motion corruption), comprehensive *in vivo* ROC evaluation of Doppler-side STAP/KA requires high-PRF raw IQ acquisitions prior to proprietary clutter suppression, with reliable metadata and task/landmark structure to define independent evaluation labels.

7 Conclusion and future work

This study presents an auditable KA-STAP framework for fUS slow-time flow detection under a tune-once-then-freeze discipline. The central practical result is that STAP is most useful when it is treated as a detector head for residual, structured nuisance rather than as a wholesale replacement

for upstream clutter suppression. Under strict low-FPR evaluation and independent structural labels where available, localized matched-subspace STAP preserves separability in regimes where conventional low-rank baselines degrade in the extreme tail, while the optional KA layer remains conservative by construction: it is shrink-only, protects a high-confidence set, and has explicit out-of-regime behavior.

The corrected SIMUS/PyMUST moving-scatterer benchmark in the Supplement should be read as a realism and fairness checkpoint rather than as a final clinical simulator. SIMUS provides a practical open RF→IQ→beamforming path and has been compared against Field II, k-Wave, FOCUS, and Verasonics in controlled settings, but it remains an approximate linear weak-scattering model and therefore requires empirical anchoring before it can stand in for a clinical acquisition model [43, 44]. Our clinically anchored *SIMUS-ClinUtility-v2* track keeps that moving-scatterer forward model, but calibrates nuisance realism to real-data telemetry before any method refreeze. The accepted nuisance profiles are now `ClinMobile-Pf-v2` and `ClinIntraOpParenchyma-Pf-v3`; the more surface-dominated intraoperative profile remains telemetry-only until a stronger real-IQ anchor can be defended.

That split is important. A first monolithic intraoperative profile could not be stabilized under the fixed hard gate, even after a deeper background redesign. The benchmark therefore now separates a competitive *parenchymal mapping* profile, where the accepted ordinary background remains brain-like and superficial high-velocity or specular structures are modeled explicitly as nuisance layers, from a development-only surface/coupling profile. This is more consistent with the clutter-filtering literature, which treats tissue background as coherent and low-rank but spatially heterogeneous enough to motivate adaptive-global and local filters [12–14], and with the intended awake-surgery and cranial-window use cases, where the parenchymal target remains central even when superficial contamination is present [8, 9, 29].

On the accepted structural *v2* benchmark, the most important result is that a frozen deployed stack closes the remaining performance gap without changing detector mathematics. A joint frozen search over the accepted development split selects RPCA→STAP with a motion-robust frozen STAP profile; on held-out accepted seeds this stack reaches $\mathrm{AUC}(\mathrm{main}, \mathrm{bg}) = 0.828$, $\mathrm{AUC}(\mathrm{main}, \mathrm{nuis}) = 0.920$, and nuisance $\mathrm{FPR@TPR}_{\mathrm{main}} = 0.5$ of 0.031. The same-residual native reference (RPCA→Kasai) gives much weaker nuisance control ($\mathrm{AUC}(\mathrm{main}, \mathrm{nuis}) = 0.442$, nuisance FPR=0.593), while the fastest strong native simple stack (HOSVD→Kasai) runs in only ≈ 0.17 s per held-out case on the RTX 4080 SUPER but again gives substantially weaker nuisance separation. The latency frontier is therefore more informative than any single runtime row. In the same accepted structural setting, a much faster STAP operating point (Adaptive Global SVD→STAP) runs in ≈ 0.34 s per held-out case while still improving all three structural summary metrics over HOSVD→Kasai. By contrast, the highest-quality accepted stack runs in ≈ 2.15 s per held-out case, with the STAP stage itself adding only ≈ 0.47 s beyond the upstream RPCA residualizer. For translational readers, the practical message is simple: RPCA→STAP is the strongest quality operating point when nuisance suppression is the priority, whereas Adaptive Global SVD→STAP is the clearest low-latency STAP operating point.

The functional checkpoint is more mixed at the full pipeline level, but it supports the same detector-head interpretation. Using only the accepted *v2* profiles and a common background-controlled GLM readout, the best end-to-end held-out pipeline remains profile-dependent: intraoperative parenchymal mapping prefers RPCA→PD-after-STAP, whereas the mobile profile prefers HOSVD→PD. These are not the same question as the detector-head audit: end-to-end

stack selection includes residualizer-profile interactions, whereas the detector-head audit isolates whether STAP improves the same residual cube under a common readout. Under that stricter stage-symmetric audit, STAP is the better detector head in 11/12 accepted family/profile comparisons, with the sole remaining miss an effective tie. In that audit the mean native-head runtime is ≈ 0.27 s per held-out functional case, the mean STAP-head runtime is ≈ 0.44 s, and the mean held-out selection-score gain is $+0.088$. Taken together, the structural and functional checkpoints support a consistent interpretation: STAP is most useful clinically as a detector head that reduces nuisance-driven errors and improves downstream map hygiene after ordinary clutter suppression.

Outside simulation, the translational evidence is supportive but intentionally not overstated. On the Macé whole-brain retrospective sweeps, label-free alias gating reduces matched-TPR false positives by 18.6% (95% CI 16.8–20.2) with a plane-wise win rate of 118/120, and the appendix-only held-out check still shows a 14.5% false-positive reduction (95% CI 7.9–21.4) with a 97.5% safety win rate across held-out planes [9, 45]. On the real-IQ ULM 7883227 bundles, same-residual detector-head audits are uniformly positive for STAP relative to both PD and Kasai (ΔAUC $\approx +0.28$ vs PD under both the `brainlike` and `elastic` motion families), which is the clearest real-data support for the detector-head interpretation [36, 37]. By contrast, the currently available Shin bundles remain mixed in the same-residual audit and should therefore be read as a low-motion real-IQ viability anchor rather than as a superiority dataset [34, 35]. For neuroscientists and neurosurgeons, this is the practical boundary of the present evidence: STAP already has a coherent nuisance-suppression and map-hygiene story on accepted benchmark regimes and selected real-IQ anchors, but it is not yet a claim of universal real-data dominance across every acquisition family.

The next steps are therefore clear. We keep `ClinIntraOpSurface-Pf-dev0` telemetry-only until a stronger surface-dominated real-IQ anchor can be defended, extend the accepted-profile functional benchmark to larger held-out splits and retrospective real-data audits, and consider detector-level changes only if a single frozen STAP head no longer remains competitive on the accepted clinically anchored benchmark. In parallel, we will continue to reduce latency via hardware-aware implementations of the batched covariance and band-energy kernels toward real-time deployment, and will explore compact frozen learned regularizers only when they preserve the shrink-only auditability and low-FPR reporting guarantees established here.

Data and code availability

All datasets used in this work are either publicly available (as cited) or derived deterministically from those sources. All code used to generate the reported tables and figures is versioned; the exact commit hash (and, when available, a public repository URL and release tag) are recorded in the reproducibility manifest (Supplementary Material). A minimal CUDA-enabled software environment is specified in `environment.yml` (Python 3.11; PyTorch 2.4 + CUDA 12.1); a container recipe is provided as `Dockerfile`. As a single-command example, Table 6 can be regenerated (pilot generation + baseline matrix + LaTeX table) by running `bashscripts/reproduce_table5_brain_kwave.sh` (expected output: `reports/brain_kwave_vnext_baselines_table.tex`).

Code repository. The reference implementation and reproduction scripts are available at `https://github.com/ArthurShune/stap-for-fus`. The canonical commit hash and (when available) a release tag are recorded in the reproducibility manifest (Supplementary Material).

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